

Strength in Numbers: Interstate Purchasing Pools and Medicaid Supplemental Rebate Capture

Jonathan Palisoc, University of Michigan, School of Public Health, Dept of Health Management & Policy

Abstract

Objective: To examine whether joining an interstate Medicaid drug purchasing pool raises supplemental rebates that states receive from drug manufacturers. Supplemental rebates are voluntary discounts negotiated above the federal statutory floor and there is minimal evidence on whether negotiating as a bloc raises rebate collections.

Data Sources: CMS-64 supplemental rebate data linked to a primary-source panel of pool membership for all 50 states, 2002–2024, reconciled from state Medicaid manuals, archived web materials, agency correspondence, and public-records requests. State Drug Utilization Data for drug-mix measures.

Study Design: Staggered difference-in-differences and event study models. Two-way fixed effects as the benchmark, with Sun-Abraham, Callaway-Sant’Anna, and de Chaisemartin-D’Haultfoeuille estimators. Wild cluster bootstrap inference.

Principal Findings: Reported supplemental rebate collections rose by 1.71 percentage points of gross Medicaid drug spending after states first entered a pool ($p = 0.013$), with heterogeneity-robust estimators at 2.13 and 2.01 pp; alternative comparison groups yield slightly smaller impacts. Outcomes also suggest certain drug purchasing pools increase rebate capture significantly more than others.

1. Introduction

1.1 Medicaid Drug Spending

Drug-related costs are a significant amount of Medicaid expenditures. After accounting for rebates, Medicaid is responsible for 11.5 percent of the \$467.0 billion spent on retail prescription drugs in the United States (CMS, 2026a). It is also the only payer operating under both statutory and supplemental rebate authority. Gross Medicaid spending on outpatient prescription drugs was \$106.4 billion in FY2024

against \$58.4 billion in rebates, leaving roughly \$48.0 billion net (MACPAC, 2026). Measured against total Medicaid spending of \$957.4 billion in FY2024, gross drug spending was about 11 percent (gross, 5 percent net) of program spending, and significantly increasing as a share in recent years (MACPAC, 2026). Federal financing is tightening at the same time, since the Congressional Budget Office (CBO) and the Joint Committee on Taxation estimate that the Medicaid provisions of the 2025 reconciliation act (P.L. 119-21) will reduce the federal deficit by \$886.8 billion over 2025-2034 (CBO, 2025).

The structure of modern pharmaceutical markets makes actually reducing drug spending challenging. Specialty and biologic drugs account for around 2 percent of prescriptions but close to 50 percent of drug spending. Unlike traditional small-molecule drugs, which eventually face generic competition that drives down prices, many biologics face limited biosimilar entry and sustained pricing power. This spending problem exists because many of these drugs work well: biologics for autoimmune conditions such as inflammatory arthritis prevent a disability that was once common, and targeted cancer drugs have changed the outlook for a number of cancers. Containing this spending without limiting access to those drugs is a common issue for state Medicaid agencies. The emergence of cell and gene therapies with per-patient costs exceeding \$1 million has introduced entirely new categories of fiscal risk. State Medicaid agencies must simultaneously contain growth in drug spending, maintain compliance with federal coverage rules, and preserve access for populations that rely heavily on public insurance. Most of the levers available to a state absorbing lower federal financing (tightening eligibility, trimming benefits, cutting provider payment) conserve dollars by reducing coverage or access. Extracting larger rebates from manufacturers is one of the few that does not limit access. Whether banding together with other states actually strengthens that lever bears directly on how states absorb a major federal Medicaid retrenchment while preserving access.

State Medicaid programs operate within a mixed federal-state structure that shapes their cost-containment options (Section 2.1). At the state level, agencies retain meaningful discretion over several levers: preferred drug list (PDL) design, prior authorization rules, managed-care pharmacy architecture, drug

utilization review, and supplemental rebate negotiations. I focus on the last of those levers: banding together with other states to negotiate as a bloc.

1.2 Medicaid Supplemental Rebates

Supplemental rebates are voluntary rebates negotiated above and beyond the federal Medicaid Drug Rebate Program (MDRP) minimum. Manufacturers pay them in exchange for preferred formulary placement, easier utilization management, or broader access to a state's covered population. As of 2019, 47 states and the District of Columbia had supplemental rebate agreements, which collectively generated over \$1.7 billion annually (OIG, 2014; Gifford et al., 2020).

Despite that fiscal importance, supplemental rebates have received very little rigorous empirical investigation. The Office of Inspector General documented substantial cross-state variation in how states collect and structure supplemental rebates but did not attempt to explain it (OIG, 2014). MACPAC (2018) estimated that supplemental rebates account for roughly 6 percent of total Medicaid drug rebates but did not investigate the drivers of variation. The academic Medicaid pharmacy literature has focused on other policies (managed care versus fee-for-service drug spending, formulary design, prior authorization, generic substitution), leaving supplemental rebates as a policy-relevant blind spot in the evidence base. In 2024, supplemental rebates ranged from near zero in some states to over 28 percent of drug spending in others. If that variation reflects replicable institutional arrangements - meaning if all states could change policy to mimic the top range of performers - pooling could represent a substantial opportunity for cost savings.

1.3 Interstate Drug Purchasing Pools

Many states attempt to increase bargaining leverage by joining interstate drug purchasing pools. The logic is straightforward: a larger covered population should strengthen the threat to steer utilization toward preferred products, which should in turn raise supplemental rebates. Three major pool families emerged in the mid-2000s: the National Medicaid Pooling Initiative (NMPI, established 2003), The Optimal PDL Solution (TOP\$, established 2005), and the Sovereign States Drug Consortium (SSDC, established 2005). By 2024, 33 of 50 states participated in one of those three arrangements.

Buyer-side market power theory suggests that larger, more coordinated purchasers can secure better prices when they can credibly redirect demand (Chown, Dranove, Garthwaite, and Keener, 2019). Structural models of formulary-rebate contracting imply that rebates depend on covered lives and on the buyer's ability to commit to preferred placement (Ho and Lee, 2024). International pooled-procurement research reaches a similar conclusion: pooling can improve procurement outcomes, but the size of the gain depends heavily on governance and implementation (Dubois, Lefouili, and Straub, 2021; Parmaksiz et al., 2022).

Pharmacy-benefit design literature indicates the same: Dranove, Ody, and Starc (2021) find that full privatization of the Medicaid pharmacy benefit decreased drug spending by 22.4 percent, with effects concentrated in states giving managed-care plans formulary flexibility. Bendicksen and Kesselheim (2022) document a countervailing trend of carve-outs, and Hernandez and Gellad (2020) show that fee-for-service programs tend to retain expensive branded drugs to capture large inflation rebates while managed-care plans adopt generics at much higher rates. On the international side, Dubois, Lefouili, and Straub (2021) find that pooled public procurement across seven low- and middle-income countries is associated with lower drug prices, with smaller reductions when the supply side is more concentrated, and Parmaksiz et al. (2022), reviewing 44 empirical studies, report savings that vary widely with the capacities of buyers, pooled-procurement organizations, and suppliers. In short, scale matters less when the buyer cannot credibly move volume.

Despite strong theory, there is no settled causal evidence on whether interstate Medicaid purchasing pools actually increase supplemental rebate capture. Existing accounts are entirely descriptive. Horvath (2019) documented that Vermont reported 4.7 percent additional savings in its first year of SSDC participation and that New York reported \$80.5 million in additional savings through NMPI, but these figures lack counterfactual comparisons.

My analysis aims to answer two questions. My first and primary aim is to measure the average change in supplemental rebate capture after a state first enters any pool using adoption-timing variation across states. The second is whether the change differs by how the pool is governed, comparing the state-

organized SSDC against the two vendor-administered pools; I treat this analysis as secondary, because pool type, adoption cohort, and the state reforms that accompany entry are difficult to separate.

To answer these questions, I construct a reconciled 50-state primary-source panel of Medicaid drug-pool membership from 2002 through 2024 and link it to federal expenditure reports. No pre-existing public dataset provides a state-year history of pool participation, and the secondary compilations that circulate disagree on a number of states, so I reconcile the record against primary and archived sources (Section 3.3) and report the coding for every state in Table 1. I find that supplemental rebate capture rises after first entry, by about 1.7 to 2.1 percentage points of gross drug spending on the state-average estimates.

2. Background and Theory

2.1 The Medicaid Drug Rebate Program

The Medicaid Drug Rebate Program was first established by the Omnibus Budget Reconciliation Act of 1990. Manufacturers enter rebate agreements with the Secretary of Health and Human Services as a condition of Medicaid coverage of their products. For brand-name drugs the basic rebate is the greater of a fixed percentage of Average Manufacturer Price (AMP) or the difference between AMP and the manufacturer's best price, plus an inflation penalty when a drug's price grows faster than CPI-U. The Affordable Care Act raised that percentage from 15.1 to 23.1 percent in 2010. For generic drugs the basic rebate is 13 percent of AMP, with an inflation-based additional rebate since 2017 (Bipartisan Budget Act of 2015, section 602).

A non-excluded commercial concession that establishes a lower best price can increase the statutory rebate, but CMS-authorized state supplemental rebates are separately negotiated and are excluded from the best-price calculation (42 C.F.R. §447.505(c)(7); Social Security Act §1927(c)(1)(C)(ii)(I)). Since 2010 the ACA has also required statutory rebates on drugs dispensed to Medicaid managed care enrollees when the plan is responsible for coverage. Extending a state's supplemental rebate agreement to

managed-care utilization remains a state option: some states collect on managed-care claims under the state agreement, others let their plans negotiate separately and keep the savings.

2.2 Preferred Drug Lists, Carve-Outs, and Uniform PDLs

Supplemental rebates are closely tied to formulary placement and utilization-management terms. A PDL is a state-published list of which drugs are “preferred” and which are not, with preferred drugs generally subject to fewer utilization-management restrictions. Preferred placement significantly increases a drug’s Medicaid market share within its class, and states use this quality to bargain steeper rebates with manufacturers.

Most literature indicates that PDLs can significantly change provider prescribing habits. Restricted-drug prescription shares fell by 9.0 percentage points in Illinois Medicaid and 6.2 percentage points in Louisiana Medicaid after PDL adoption (Virabhak and Shinogle, 2005). Florida’s state-mandated PDL reduced overall drug use by 6 percent while increasing plan costs by 27 percent, because the list channeled utilization toward preferred branded drugs with high rebate yields (Munshi et al., 2018). A PDL that maximizes supplemental rebate revenue may therefore not minimize net drug costs, which is one reason this paper measures rebate capture instead of net spending.

Two additional architectural choices may affect how much of a state’s drug volume a PDL moves. The first is whether the pharmacy benefit is carved out of managed care, meaning the state administers it directly through fee-for-service (FFS) and runs one PDL for the whole benefit, or carved in, meaning managed-care organizations (MCOs) administer it for their enrolled members. The second is whether the state binds those MCOs to a uniform PDL; Exhibit 1 assigns every state to one of the four combinations as of 2024.

Exhibit 1. Pharmacy carve-out status by uniform PDL status, 2024

	Uniform PDL	No uniform PDL
Pharmacy carved out of	19 states: AL, AK, AR, CA, CT, ID, ME,	1 state: SD

managed care	MS, MO, MT, NY, ND, OH, RI, TN, VT, WV, WI, WY	
Pharmacy carved in to managed care	15 states: DE, FL, IL, IA, KS, KY, MI, MN, NE, NC, PA, SC, TX, VA, WA	15 states: AZ, CO, GA, HI, IN, LA, MD, MA, NV, NH, NJ, NM, OK, OR, UT

Notes: State pharmacy carve-out and uniform PDL status as of 2024, from the pharmacy-benefit structure panel described in Section 3.4, and summing to 50 states. This is a descriptive snapshot of where states stand now, and not a measure of the architecture in place when states entered pools, most of which entered before 2010. Author's coding from state Medicaid manuals, managed-care contracts, and archived state materials.

Carving pharmacy out of managed care effectively delivers a single statewide list. A pool combines several states' covered lives into one negotiation, but each member state still makes its own PDL decision, so what the pool assembles is the sum of what its members can individually steer. Even though each state's PDL is different, pooling provides a joint negotiation calendar, shared clinical review, and manufacturers bidding to the bloc, with each state then accepting or declining terms for itself.

2.3 Pool Organization and Empirical Predictions

The three pool families differ in who owns the rebate contracts, who runs the negotiation, and who decides whether to take the terms. SSDC describes itself as the only Medicaid rebate pool organized and managed by the states. Member states contract individually with manufacturers using their own supplemental rebate agreements, while negotiating with the consortium's combined covered lives behind them, and each state decides for itself, on an annual member-state cycle, whether to accept what the negotiation produced. NMPI and TOP\$ are vendor-administered, so a commercial administrator negotiates collective rebate agreements on the member states' behalf and owns the contract paper. In short, the contrast the paper examines is who owns the bargaining. Further institutional detail is in Appendix B.

Pooling may increase bargaining leverage when member states can credibly coordinate formulary decisions. The logic is as follows: What a manufacturer will pay for preferred placement depends on what it stands to lose without it, which is the volume a formulary decision can credibly move away from its drug times the margin on that volume. A state bargaining alone puts only its own steerable volume behind

the threat of nonpreferred placement. A pool puts the combined bloc's volume behind the same threat, which worsens the manufacturer's no-deal outcome and raises the rebate the states can extract.

Three empirical implications follow, which I present as my main hypotheses: First, size alone is not enough: the threat is only as good as the volume the pool can actually steer, so a pool that assembles nominal enrollment but cannot coordinate formulary action, because PDL decisions are fragmented across plans running their own lists, adds covered lives without adding bargaining power. Pooling and steerable volume are complements, and the within-state arrangements that produce steerable volume are the ones Exhibit 1 measures. Second, smaller states should gain more from pooling. A small state bargaining alone has little volume to threaten with, so pooling relaxes the constraint exactly where it binds hardest (Section 5.6). Third, pool governance may matter. SSDC is state-owned, so the parties who keep the rebate savings also control the bargaining, while a vendor administrator's objectives, such as client retention and standardized operations across states, need not rise and fall with each incremental rebate dollar. The features that should make a state-run pool effective also shape who selects into it, and pool entry often arrives together with exactly those reforms, so part of a measured pool effect may reflect the correlated bundle.

3. Data

3.1 Supplemental Rebate Capture

The primary outcome is the supplemental rebate capture rate, defined as: reported supplemental rebate collections as a share of gross Medicaid drug spending. Both the numerator and the denominator come from CMS-64 Financial Management Reports, the quarterly expenditure reports states file with CMS to draw federal financial participation. The numerator is the state supplemental (sidebar) rebate collection line, which has appeared on the form since 1998 and which states report separately from the national statutory MDRP rebate; from 2010 it also includes the managed-care counterpart line. The denominator is

total reported drug expenditures. I use a share because states differ enormously in program size, so the outcome measures whether a state collects more relative to its own drug program.

The numerator covers every supplemental rebate a state reports, including agreements negotiated outside any interstate pool. A state may hold single-state, pooled, value-based, and managed-care supplemental agreements at the same time, and the CMS-64 does not separate them. An increase in the numerator after pool entry is therefore an increase in total reported state supplemental collections, and attributing it specifically to bloc bargaining requires an assumption these data cannot verify (Section 6.3).

In addition, the numerator and the denominator are not in the accounting sense. As MACPAC notes, the time required to collect utilization information and invoice manufacturers means the rebates collected in any particular quarter are generally attributable to drugs purchased in prior quarters. Section 5.3 therefore reports the ratio built on a lagged denominator and on two-year moving sums.

Building a consistent panel requires harmonizing several reporting formats across years, and I also rebuild the outcome from the per-fiscal-year sheets of the CMS-64 workbook as a check (Appendix A). After standardization the panel contains 1,150 state-years from 2002 through 2024. The CMS-64 also reports on a fiscal-year basis, which need not align with calendar-year membership indicators (Section 6.3).

3.2 State Drug Utilization Data

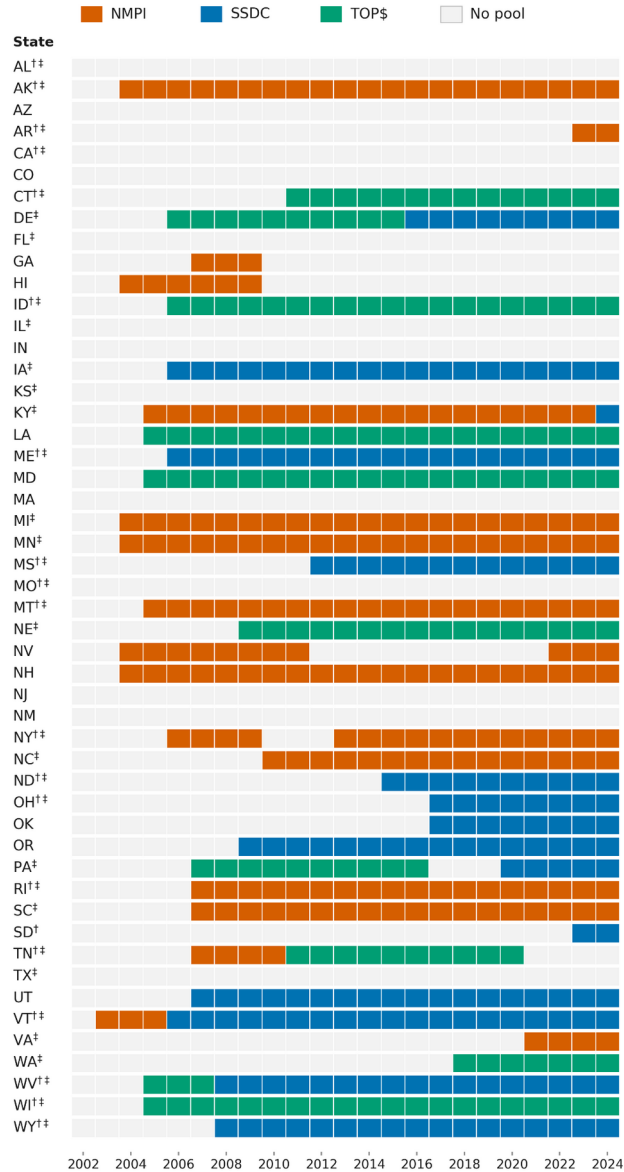
State Drug Utilization Data (SDUD) are the drug-level utilization reports states file with CMS for rebate invoicing, at the state, quarter, and national drug code (NDC) level. I aggregate them to the state-year level and use them for two purposes: First, I cross-check total drug spending against the CMS-64 denominator. Second, it allows building drug-mix and concentration measures the CMS-64 does not contain: the share of spending on single-source (protected brand) products, the share on innovator multiple-source products, average spending per prescription, and the share of spending accounted for by the top ten NDCs. These support the composition and steering checks in Section 5.6. The SDUD cannot

replicate the main results, because it contains no rebate data and has category crosswalk issues, so I mainly report related findings as informational.

3.3 Pool Membership

I construct the first novel policy panel that documents pool membership for all 50 states from 2002 through 2024. Because no reconciled public history exists, I built it from state Medicaid agency materials including archived content, direct contact with each state's MDRP contacts, and public-records requests. Where possible, I validate against KFF State Health Facts, Horvath (2019), OIG (2014), and CMS supplemental-rebate-agreement tables, with each contested state resolved against archived sources. Pool indicators are mutually exclusive within a state-year: each state-year is coded SSDC, NMPI, TOP\$, or no pool. Five states switch from a vendor pool into SSDC during the panel (Delaware, Kentucky, Vermont, West Virginia, and, after a no-pool gap, Pennsylvania). SSDC membership is absorbing, since no state has left it, while NMPI and TOP\$ membership reverses for several states. Table 1 reports the coding.

Table 1. State Pool Membership and Pharmacy-Benefit Coding, 2002-2024



Notes: One row per state, in alphabetical order. Each cell shows the state's pool membership in that year: colored cells mark NMPI, SSDC, or TOP\$ membership per the legend, and light gray cells mark years with no pool membership, so first entries, exits, gaps, re-entries, and switches between pools are visible directly. † marks states with a pharmacy carve-out and ‡ marks states with a uniform PDL, both as of 2024. States ever in a pool form the treated group and states never in any pool form the comparison group; Alaska, although in NMPI throughout, is excluded because it reports no supplemental rebate values. Tennessee's sequence is provisional, and ten further states carry narrower dating caveats listed in Appendix C. Author's coding from primary-source state Medicaid materials, CMS supplemental-rebate-agreement tables, and public-records requests.

3.4 Pharmacy-Benefit Structure Panel

A second state-year policy panel measures pharmacy-benefit structure directly, with 747 coded spells across all 50 states covering uniform PDL status, pharmacy carve-out status, MCO formulary autonomy, PDL and PBM vendor, rebate-negotiation model, and major reform flags. Its governance variables are

densely coded only from 2010 onward, with roughly 2 percent non-missing before 2010 against 92 to 100 percent from 2010, so for most pool entries before 2010 the panel cannot say whether a governance reform preceded or followed pool entry. Governance variables therefore never enter the primary specification and appear only as a sensitivity check on the post-2010 window (Appendix D), while the reform audit in Section 5.3 handles the bundling question at the level of named states and dated reforms. Carve-out and uniform PDL status in the most recent coded year, shown in Exhibit 1 and Table 1, are observed for all 50 states.

3.5 Analytic Sample and Comparison Group

The estimation panel is the full 2002-2024 panel of 49 states with outcome data, of which 36 ever pooled are defined as treated, across 15 adoption cohorts spread over 2003 to 2023.

That never-pooled comparison group is small and significantly heterogenous compared to the average US state, which may indicate some level of selection. It includes four of the six most populous states that plausibly found little value in pooling due to already large size (California, Texas, Florida, and Illinois), alongside small programs.

Table 2 reports the baseline characteristics group by group.

Table 2. Pre-Entry Baseline Characteristics by Ultimate Pool Group

Group	States	State-years	Mean capture (%)	Median capture (%)	State-years at zero (%)	Mean log gross drug spending
Never-pooled	13	287	2.40	1.42	37.3	20.72
Vendor pool only (NMPI or TOP\$)	21	130	1.02	0.00	50.8	20.10
SSDC early adopters (2006-2009)	7	30	0.59	0.00	80.0	18.80

SSDC later adopters (2012+)	8	84	1.11	0.00	67.9	19.35
-----------------------------	---	----	------	------	------	-------

Notes: Capture is supplemental rebate collections reported on the CMS-64, as a percentage of gross Medicaid drug expenditures, and means and medians pool state-years within each group. Each treated state contributes its own pre-entry years, meaning all years before its first entry into any pool, and the never-pooled states contribute all panel years, the same convention as Figure 2. Groups are fixed by ultimate 2002-2024 membership, with the SSDC states split at the 2012 adoption cutoff; the five states that entered SSDC from a vendor pool are grouped by their SSDC cohort, and their baseline years precede their first entry into any pool. “State-years at zero” is the percentage of state-years in the window reporting exactly zero supplemental collections. Gross drug spending is missing for seven pre-entry state-years, which are omitted from the log-spending mean. Alaska is excluded because it reports no supplemental rebate values. These are descriptive comparisons and are not causal. Author’s calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

SSDC adoption is also examined separately, since it is the one pool with an absorbing treatment. Of the fifteen states that adopted SSDC through 2024, seven are early adopters first treated between 2006 and 2009 (Iowa, Maine, and Vermont in 2006; Utah in 2007; West Virginia and Wyoming in 2008; Oregon in 2009). Eight are later adopters first treated in 2012 or later (Mississippi 2012, North Dakota 2015, Delaware 2016, Ohio and Oklahoma 2017, Pennsylvania 2020, South Dakota 2023, and Kentucky 2024). The early adopters were predominantly smaller, rural states, while the later wave included Pennsylvania, one of the largest state Medicaid programs.

4. Empirical Strategy

4.1 Treatment Definitions and Estimands

I use staggered difference-in-differences (DID) as my main design. States that entered a pool are considered as the “treated” group and the thirteen states that never joined any pool serve as “untreated” comparisons. The design compares the change in supplemental rebate capture within entering states, before versus after entry, with the change over the same years in the never-pooled states. State fixed effects remove permanent differences across states, and year effects remove national shocks that hit all states together, such as the post-2010 escalation of federal statutory rebates.

Pool membership is not one quantity once states leave and switch pools, so I fix four treatment definitions in advance (Table 3). The first-entry definition is absorbing: the indicator turns on in the first year a state belongs to any pool and stays on, even if the state later leaves. That is the definition the Sun-Abraham and Callaway-Sant’Anna estimators accept, and it describes the change following first entry. The change from

currently being in a pool is a separate quantity, captured by the co-primary current-membership definition, which turns the indicator off at exit. The third definition, continuous membership, keeps the indicator on from first entry but censors a state's observations after its first exit. The fourth splits current membership into separate indicators for SSDC, NMPI, and TOP\$, and it exists for the governance comparison in Section 5.4.

Table 3. Treatment Definitions

Definition	Indicator behavior	Estimators	Role
First entry, absorbing	On from first entry into any pool, stays on after a later exit	TWFE, Sun-Abraham, Callaway-Sant'Anna	Primary; effect following first entry
Current membership	On only in years the state is in a pool	TWFE, dCDH	Co-primary; effect of current membership
Continuous membership	On from first entry, with observations censored after the first exit	TWFE, Sun-Abraham, Callaway-Sant'Anna	Sensitivity
Specific pool	Separate current-membership indicators for SSDC, NMPI, and TOP\$	TWFE, dCDH	Governance comparison

Notes: TWFE is two-way fixed effects; dCDH is the de Chaisemartin and D'Haultfoeuille (2020) estimator, which allows the treatment indicator to switch off and is therefore the preferred estimate for current membership, with the TWFE value a benchmark that weights and compares differently. Sun-Abraham and Callaway-Sant'Anna require an absorbing treatment and are not estimated for the definitions that reverse. The comparison group is the same 13 never-pooled states under every definition.

The primary estimand is the total effect of first entering an interstate purchasing pool on reported supplemental rebate capture. This includes implementation and governance changes induced by participation, whether they occur at entry or subsequently. Independent policy changes are not part of that causal effect merely because they occur after entry; if they generate differential outcome trends, they remain a threat to identification. Within SSDC, the seven states that entered between 2006 and 2009 have four to seven observed pre-entry years. Those pre-periods are unequal and their post-periods largely correspond to a smaller consortium, so Section 5.5 reports the later-adopter estimate alongside the all-adopter SSDC estimate.

4.2 Adjusting for Post-Entry Governance Changes and Other Covariates

The primary specification carries no covariates for two reasons. Conceptually, the candidate controls, meaning uniform PDL status, carve-out status, and the negotiation model, are the channels through which pooling is supposed to work, since states unify their PDLs or extend rebate collection because they pooled - or in other words, mechanisms through which pooling affects reported collections rather than pre-treatment covariates. Conditioning on a treatment-induced change can remove part of the effect of interest and introduce post-treatment bias. As the intent of my main estimation is to solely estimate the total effect of first pool entry, *including subsequent changes induced by participation*, the primary models include state and year fixed effects but do not condition on contemporaneous pharmacy-benefit governance. Governance-controlled models are therefore a uniform sensitivity check reported in Appendix D, and the design addresses the bundling concern at the level of named states and dated reforms (Section 5.3) instead of by regression adjustment.

4.3 Estimation and Inference

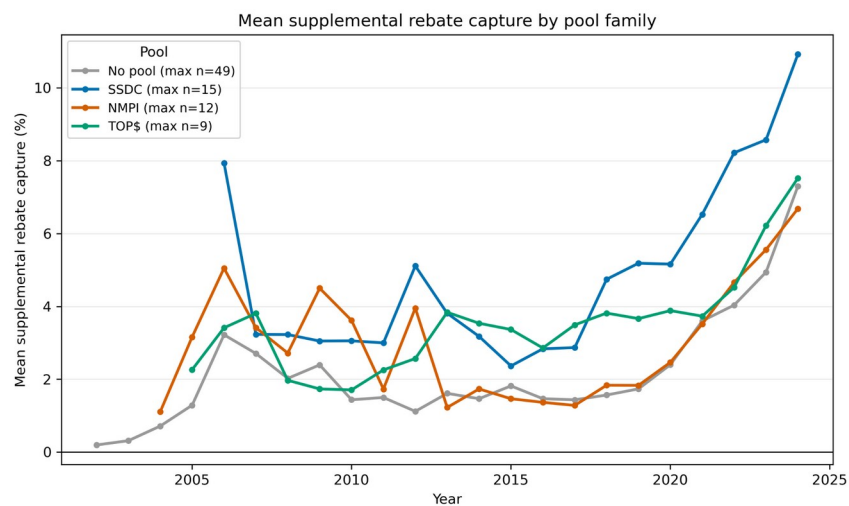
I report TWFE and heterogeneity-robust (Sun and Abraham (2021), Callaway and Sant’Anna (2021) de Chaisemartin and D’Haultfoeuille (2020)) treatment effects. To avoid overstating significance with so few state clusters - 49 in the estimation panel and only 13 never-pooled comparison states - the main estimate and inference uses the wild cluster bootstrap with Rademacher weights and 9,999 replications, clustered at the state level. I follow Rambachan and Roth (2023) sensitivity framework to the primary any-pool event study to account for any pre-trend absorption. All estimates weight states equally unless noted, and Section 5.3 reports the gross-spending-weighted version alongside them. Appendix G carries the estimating equations, the remaining inference procedures for the SSDC later-adopter subgroup, the Goodman-Bacon (2021) diagnosis of the two-way fixed effects benchmark, two further implementations, and a replication of every specification in a second statistical package.

5. Results

5.1 Raw Data

Pooled states capture more than never-pooled states today and captured less before they joined. Figure 1 plots the path of mean supplemental rebate capture by pool family over time. SSDC state-years sit above never-pooled state-years in most years after 2006, with the vendor pools in between.

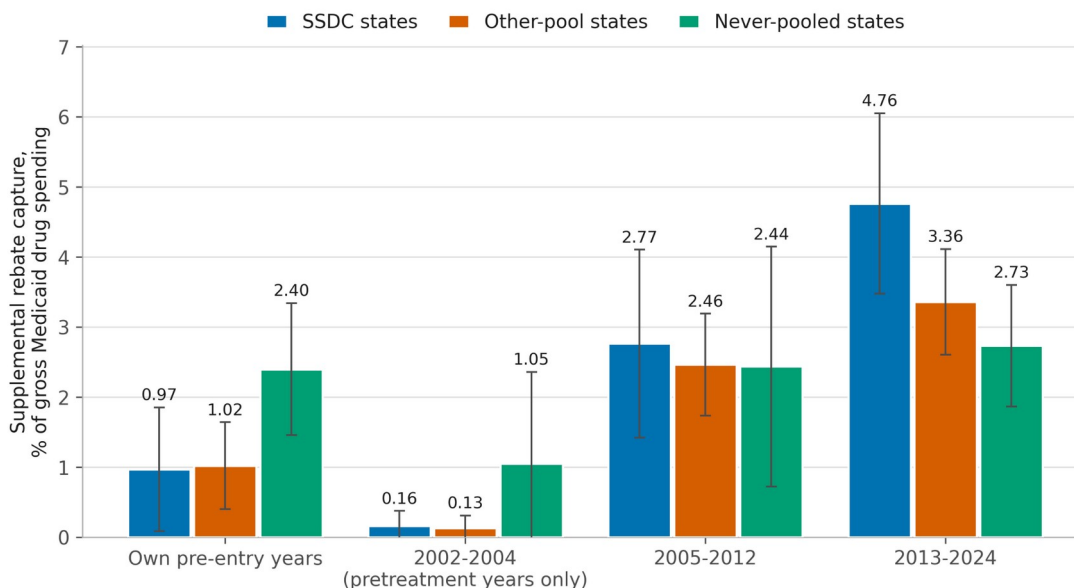
Figure 1. Mean supplemental rebate capture by calendar year and pool family



Notes: Unweighted mean of the supplemental rebate capture rate, supplemental rebate collections reported on the CMS-64 as a share of gross drug spending, across states in each pool family in each year, using membership, so group composition changes over time. Author's calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

Figure 2 plots mean capture for three fixed groups of states: 15 that ever entered SSDC, the 21 that ever entered a vendor pool but never SSDC, and the 13 that never pooled. 2002 is the only wholly pretreatment year as Vermont entered NMPI in 2003 and six more states entered in 2004, so I report both a 2002-2004 window and each state's mean over its own pre-entry years.

Figure 2. Supplemental rebate capture by period and ultimate pool membership



Notes: Bars are the mean capture rate across state-years, supplemental rebate collections reported on the CMS-64 as a percentage of gross Medicaid drug spending, with 95 percent confidence intervals from a state-clustered standard error. Groups are fixed by ultimate membership over 2002-2024: 15 ever-SSDC states, 21 ever-other-pool states, and 13 never-pooled states. “Own pre-entry years” uses each state’s years before its first pool entry, and all panel years for the never-pooled states. Author’s calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

States that later pooled started from below-average capture, indicating some awareness of relative underperformance. Over their own pre-entry years the future SSDC states averaged 0.97 percent of drug spending and the future vendor-pool states 1.02 percent, against 2.40 percent for the states that never pooled. Interestingly, by 2013-2024 the ordering had reversed, at 4.76, 3.36, and 2.73 percent. Low baseline capture is equally consistent with mean reversion, which Section 6.3 keeps as a live alternative. Raw capture rates by pool family are in Table A1.

5.2 Main results

Rebate capture rises after entry under every estimator. Table 4 reports the primary estimates: states that entered a pool saw supplemental rebate capture rise by 1.71 percentage points of gross drug spending under the static two-way fixed effects specification (clustered SE 0.66), with a wild cluster bootstrap p of 0.013 and a bootstrap confidence interval of [0.35, 3.11]. The heterogeneity-robust estimators land in the same place, at 2.13 pp for Sun-Abraham and 2.01 pp for Callaway-Sant’Anna.

Table 4. Effect of Pool Entry on Supplemental Rebate Capture, 2002-2024

	Two-way fixed effects	Sun-Abraham	Callaway-Sant'Anna	dCDH
Panel A. First entry, absorbing (primary)				
Effect (pp)	1.711 (0.664)	2.131 (1.005)	2.006 (0.906)	-
95% confidence interval	[0.35, 3.11]	[0.16, 4.10]	[0.23, 3.78]	-
Wild cluster bootstrap p	0.013	-	-	-
Common window, event times 0-8 (pp)	1.707	1.866	1.813	-
Panel B. Current membership (co-primary)				
Effect (pp)	1.826 (0.588)	-	-	1.218 (0.559)
95% confidence interval	[0.59, 3.07]	-	-	[0.12, 2.31]
Wild cluster bootstrap p	0.004	-	-	-
Panel C. Continuous membership (sensitivity)				
Effect (pp)	1.673 (0.663)	2.252** (0.940)	2.096** (0.921)	-
Wild cluster bootstrap p	0.015	-	-	-
Observations	1,111	1,111	1,111	1,111

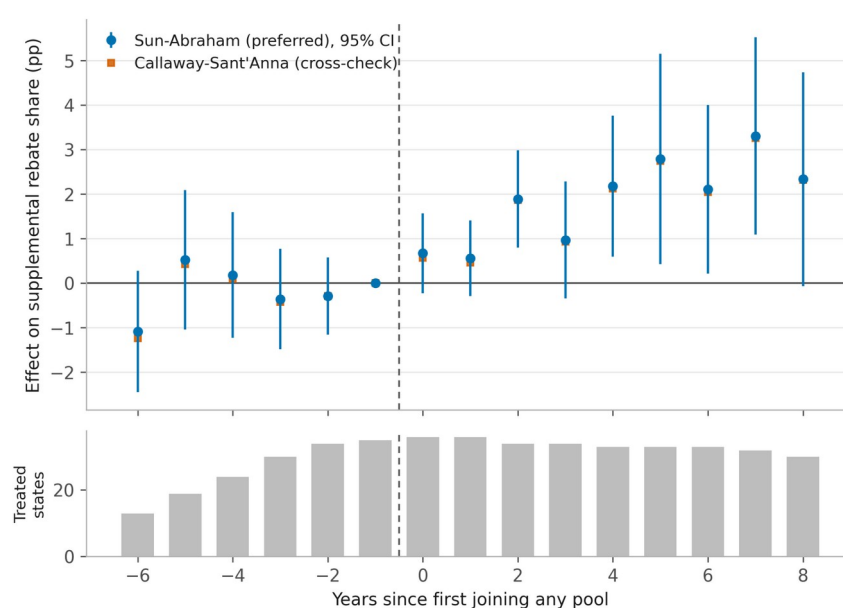
Notes: The outcome is the supplemental rebate capture rate, supplemental rebate collections reported on the CMS-64 as a percentage of gross Medicaid drug expenditures. Standard errors clustered on state are in parentheses. All specifications include state and year fixed effects and no covariates, weight states equally, and compare the 36 ever-pooled states with the same 13 never-pooled states; treatment definitions are those in Table 3, and the drug-dollar-weighted estimand is in Table 5, Panel D. The common-window row restricts the first-entry estimates to event times 0 through 8. The Observations row counts the 2002-2024 estimation panel; the continuous-membership specification in Panel C censors state-years after a first exit, leaving 1,043 (Table B5). The two-way fixed effects intervals are wild cluster bootstrap intervals; the Sun-Abraham and Callaway-Sant'Anna intervals are analytic and overstate significance with this many clusters. For current membership, dCDH is the preferred estimate because it accommodates exit. The Sun-Abraham estimates supersede earlier values produced by a software implementation that excluded the single 2003-entry cohort from the estimate while retaining its post-entry years as comparison observations; Appendix G documents the correction. Callaway-Sant'Anna uses the never-treated group with the simple aggregation, and Alaska is excluded because it reports no supplemental rebate values. Stars appear only on the Panel C Sun-Abraham and Callaway-Sant'Anna estimates, whose panel reports no interval or bootstrap p; they are based on the analytic cluster-robust p-values and, like the analytic intervals, overstate significance with 13 never-pooled comparison states. *p<0.10, **p<0.05, ***p<0.01. Author's calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

Effects of current pool membership are lower at 1.22 pp [0.12, 2.31] under dCDH but I am unable to determine whether the gap between the two reflects exit effects, weighting, or carryover.

Figure 3 plots event times -6 through +8. The pre-entry path is flat, with a Sun-Abraham pre-period coefficient averaging -0.67 pp then rising and persisting to about 2 to 3 pp by event years 4-8.

The lower panel of Figure 3 reports how many treated states identify each event time. Support is thinnest in the early leads, which rest on as few as eight treated states in the full-range version of the figure, although 29 of the 36 treated states are still observed twelve years after entry. The long lags describe the earlier-adopting cohorts, since reaching event time twelve in a panel ending in 2024 requires joining by 2012 and are estimated less precisely. The Sun-Abraham standard error rises from 1.226 pp at event time eight to 1.465 pp at event time twelve, and past event time nine every interval contains zero (Figure B1).

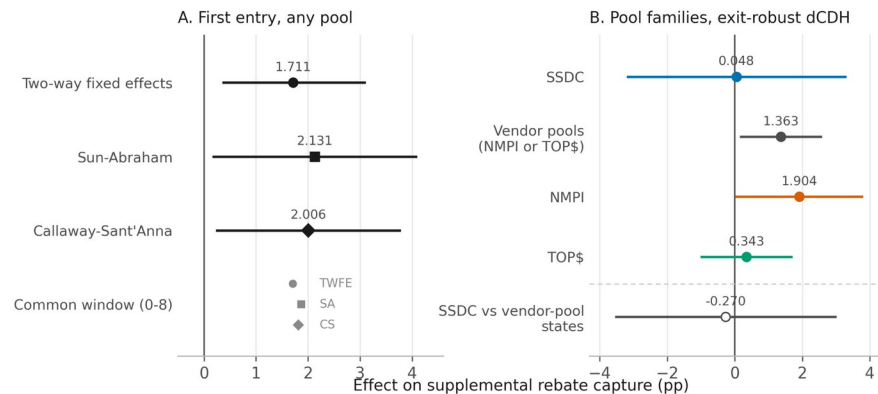
Figure 3. Any-pool event study, supplemental rebate capture



Notes: Event-time coefficients from the Sun-Abraham interaction-weighted event study with 95 percent confidence intervals, with Callaway-Sant'Anna estimates overlaid as a cross-check and visually indistinguishable from them. Event time is years since a state first entered any pool, event time -1 is the omitted reference, and the lower panel reports the number of treated states contributing at each event time. The window is restricted to six years before through eight years after first entry, because the earliest leads rest on the fewest treated states and event times past +8 are identified by an increasingly early-adopting group of states and estimated less precisely. Figure B1 reports the full range. Author's calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

For scale, the never-pooled states captured about 2.7 percent of gross drug spending in supplemental rebates in 2013-2024. Figure 4 collects the first-entry estimates from Table 4 and the pool-family estimates from Table B1. The first-entry estimates sit between 1.7 and 2.1 pp whichever estimator produces them, while the pool-family intervals are wider, overlap one another, and cross zero in three of the five rows.

Figure 4. First-entry estimates by estimator and pool-family estimates, supplemental rebate capture



Notes: Panel A plots the first-entry estimates from Panel A of Table 4 with 95 percent confidence intervals. The two-way fixed effects interval is the wild cluster bootstrap interval; the Sun-Abraham and Callaway-Sant'Anna intervals are analytic and overstate significance with 13 never-pooled comparison states. The Sun-Abraham estimate is the corrected value printed in Table 4, which supersedes the earlier implementation documented in Appendix G. The common-window points restrict each estimator to event times 0 through 8, and Table 4 reports them as point estimates, so they are plotted without intervals. Panel B plots the pool-family estimates from Table B1 under the exit-robust dCDH estimator, the only column of that table reported with bootstrapped 95 percent confidence intervals. The intervals are wide and overlap across families, which is why Section 5.4 does not rank the pool families. The bottom row is the direct contrast of the 15 ever-SSDC states against the 21 ever-vendor-pool states, so its comparison group differs from the never-pooled comparison used in every other row. Author's calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

5.3 Sensitivity

Table 5 reports sensitivity checks. Under alternative accounting conventions, treatment definitions, reweighted and synthetic comparison groups, and single-state exclusions, the estimate stays between roughly 1.1 and 2.1 percentage points.

Table 5. Any-pool Sensitivity

Specification	Two-way fixed effects (pp)	WCB p	Heterogeneity-robust (pp)	N
Panel A. Treatment definition				
Primary: first entry, absorbing	1.711	0.013	SA 2.131; CS 2.006	1,111
Current membership	1.826	0.004	dCDH 1.218	1,111
Panel B. Outcome and accounting				
Prior-year gross spending as denominator	1.626	0.026	-	1,055
Fee-for-service sidebar numerator only	1.914	0.016	-	1,104
2010-2024 window only	0.887	0.611	-	720
Panel C. Counterfactual				

construction				
Entropy-balanced comparison group	1.490	0.019	SA 1.522	1,089
Synthetic DID, 2002-2024 panel	1.880	-	SC 1.346	1,127
Same-year observed-reform states dropped	1.897	0.006	SA 2.187	1,088
Tennessee dropped	1.592	0.021	SA 1.926	1,088
Panel D. Target-population weighting				
Gross-spending-weighted	3.589	0.227	SA 8.246	1,111

Notes: The outcome is the supplemental rebate capture rate, supplemental rebate collections reported on the CMS-64 as a percentage of gross Medicaid drug expenditures, unless the row changes the denominator. Panels A through C report average state estimands and Panel D reports the drug-dollar-weighted estimand. “WCB p” is the wild cluster bootstrap p-value for the two-way fixed effects estimate. SA is Sun-Abraham, CS is Callaway-Sant’Anna, and SC is synthetic control. Sun-Abraham values reflect the implementation correction documented in Appendix G. Synthetic estimates are aggregated across treated post-treatment cells and have no bootstrap analogue. On the 2002-2024 panel 13 of the 15 cohorts have enough pre-treatment years to be given a synthetic control, and sixteen missing outcome cells are linearly interpolated within state for that family only. The gross-spending weight is each state’s mean gross drug spending over its own pre-entry years, so the weight is not itself treated. Author’s calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

As an additional check on the rebate-collection lag, I divide collections by prior-year rather than same-year gross spending and rebuild the outcome as a two-year moving ratio, which gives 1.63 and 1.52 pp.

Restricting the panel to 2010-2024 - the years under the higher statutory floor - reduces the estimate to 0.89 pp with a bootstrap p of 0.61, indicating a rising floor may reduce supplemental collections.

Reweighting the never-pooled states by entropy balancing or propensity scores pulls the estimate down to 1.37 to 1.52 pp (Appendix B), although with 12 usable donor states the reweighting has limited room to work. Synthetic-control estimation gives 1.09 to 1.88 pp and dropping states gives 1.59 to 1.99 pp. First entry, averaged across states, clusters at 1.7 to 2.1 percentage points, current membership runs 1.22 to 1.83, and the alternative counterfactuals run 1.09 to 1.88. Overall, the state-average estimates span roughly 1.1 to 2.1 points, each attached to the treatment definition and counterfactual that produced it.

One estimand behaves differently: weighting states by their own pre-entry gross drug spending gives 3.59 pp under two-way fixed effects (bootstrap p = 0.227) and 8.25 pp under Sun-Abraham, with an interval of [1.32, 15.18].

I note issues with differential trends. I follow Rambachan-Roth and find that the estimate holds up only if differential trends stay under about 10 to 15 percent of the largest pre-entry deviation. Under the

smoothness restriction the confidence set contains zero once second differences are allowed to change by 0.1 percentage points a year.

5.4 Difference between pools

A TWFE direct contrast comparing the 15 ever-SSDC states against the 21 ever-vendor-pool, never-SSDC find a different of +2.08 pp ($p = 0.061$) but falls to an insignificant -0.27 pp under alternative Sun-Abraham and Callaway-Sant'Anna estimations. As such, I do not claim a causal governance advantage to any one pool (Figure 4, Panel B, and Tables B1 and B2).

5.5 Later SSDC Entrants

The eight states that entered SSDC in 2012 or later are estimated to increase rebate capture by 3.38-3.71 pp, but the estimation violates parallel trends. The all-adopter SSDC estimate that pools in the seven early entrants is a null 0.61 pp.

5.6 Secondary Outcomes, Mechanism, and Heterogeneity

As an additional check, I run the later-adopter estimator on other outcomes built from the same data, which separate capture from savings (Table E1). Supplemental rebate dollars per prescription rise \$3.19 alongside the 3.38 pp capture gain. However, I find no detectable change in log gross drug spending (-0.04, SE 0.05), so this does not establish lower net spending. Appendix J explains why the directly reported all-rebate net series cannot be read as a pool-specific effect here. The within-state top-10 NDC spend share rises 2.18 pp (SE 0.69), which is consistent with formulary steering, although aggregate state-year data cannot separate steering from changes in drug mix, prices, or high-cost product launches.

6. Discussion

6.1 Interpretation and Policy Implications

Supplemental rebate capture rose by 1.71 percentage points of gross Medicaid drug spending after states first entered a pool, with heterogeneity-robust estimators at 2.13 and 2.01 pp. Definitions that turn treatment off when states exit give 1.22 to 1.83 pp, and the reweighted and synthetic comparison groups show slightly smaller effects, at 1.09 to 1.88 pp. The drug-dollar-weighted estimates are larger and not distinguishable from zero. The event study is flat before entry and rises over several years with no jump in the join year. It is unclear whether the state-organized consortium does better than the vendor-administered pools.

The estimates translate into dollars only under strong assumptions about generalizability, existing pool participation, and federal-state sharing. The state-average and drug-dollar-weighted estimands differ by a factor of two to four, and the drug-dollar-weighted version is not distinguishable from zero, so a national dollar figure would be far less certain than the state-average estimates suggest. Most states that have not pooled are also large programs that plausibly already negotiate at scale, and they are the states for which the estimated average is least likely to transfer. The pool-family estimates flip sign across estimators (Section 5.4), so I cannot rank the three arrangements, and a state deciding which pool to join gets no guidance from these estimates.

Extracting a rebate is itself a reduction in payment, and the difference is in what the payer assumes about the response. When a state cuts provider rates, physicians and hospitals can decline to serve Medicaid patients, and access falls directly, whereas supplemental rebate instead presumes a deal is struck and the drug stays available. That presumption is reasonable for drugs that win preferred placement and weaker for the ones that lose it, since a rebate-maximizing PDL restricts nonpreferred products through prior authorization and step therapy. Pooling can raise rebate capture without directly reducing eligibility or

provider fee schedules. Pool-supported PDL decisions may still affect prescribing and access, which this paper does not measure.

While a gain of one to two points of gross drug spending may seem relatively small, it can still provide significant benefits to programs and its beneficiaries. When federal financing falls, a state that collects more from manufacturers has less reason to narrow eligibility, restrict the drugs it covers, or cut the benefit itself. Even a modest rebate gain therefore has a clinical side, since it helps keep biologics for autoimmune disease and targeted cancer drugs available to the patients who rely on Medicaid for them (Section 1.1).

6.2 Relation to Prior Studies

The finding lines up with the buyer-power and pooled-procurement literatures. Chown et al. (2019) and Ho and Lee (2024) imply that scale raises rebates only when attached to a credible formulary decision, and the concentration result in Section 5.6 is an indirect Medicaid-specific version of that pattern. The international evidence that pool design drives outcomes (Dubois et al., 2021; Parmaksiz et al., 2022) is consistent with the size of the average change found here. I add to the literature by providing a counterfactual estimate for the self-reported savings figures in Horvath (2019); magnitudes are similar.

6.3 Limitations

My study design has several strengths: a full 2002-2024 panel, an outcome measured from mandatory federal expenditure reports instead of self-reported savings, and a comparison group held fixed across every specification. Potential limitations include threats to identification, measurement error, and limited external validity.

Due to data quality, endogeneity of pool entry, and prevalent co-timing of other related policies, I interpret my estimates narrowly as “reported capture” rather than savings from the pool’s collective bargaining. An increase of one to two points against a never-pooled base of under three percent is a large proportional improvement for a state currently reporting almost no supplemental rebate collections, and a

small amount against Medicaid drug-spending growth even if every dollar were an additional net collection. These data also measure neither the state share of the gain nor the administrative cost of collecting it.

6.3.1 Identification

Identification turns on how much post-entry differential movement the estimate can absorb, which through Rambachan-Roth I find as minimal. Because pooling states started from below-average baseline capture, mean reversion among low-baseline states is a live alternative. A group reverting to the mean should already be drifting up before entry, and the flat pre-entry path in Figure 3 argues against the simplest version of that story. However, the design cannot rule it out. The comparison group is small and selected on program size and negotiating capacity, and reweighting the never-pooled states toward the treated states' baseline profile lowers the estimate by 0.2 to 0.8 percentage points to between 1.37 and 1.52 (Appendix B). Pool entry can co-time with other pharmacy-benefit reforms, so the estimate reflects pool participation as implemented, and the audit under-detects those reforms for states entering before 2010. Adoption is policy-endogenous and treatment timing is measured with some uncertainty for a few states (Appendix C).

6.3.2 Measurement

The numerator is every supplemental rebate a state reports and not pool-attributable rebates, so an increase after entry cannot be assigned to bloc bargaining without an assumption these data cannot verify. In addition, about two-thirds of treated-state pre-entry years report exactly zero collections against about one-fifth after entry, so an unknown share of the estimated change is states beginning to collect, or beginning to report, rather than a deeper rebate on volume they were already collecting. To limit the influence of the remaining reporting problems I 1) validate the CMS-64 outcome against the SDUD and rebuild it from the underlying line items, and 2) re-estimate with negatives dropped and with outliers winsorized.

6.3.3 External validity and policy interpretation

The later-adopter estimate is identified off eight states that entered a large, established pool, so it does not describe the early cohorts. The remaining non-adopters may not resemble prior adopters either, which limits how well these estimates predict the change for the next state to enter. Rebate capture is also not net drug spending, since a rebate-maximizing formulary can steer utilization toward high-rebate branded drugs, and the access and prescribing consequences of pool-negotiated formularies are not measured here. Precision is limited mainly by the small never-pooled comparison group, so several confidence intervals hug zero and the heterogeneity cuts are underpowered.

6.4 Future Research

Future research should further examine the exact functionality of pools, information on which is currently opaque. Class-level mechanism evidence, which would require class-coded rebate data joined to a therapeutic-class crosswalk, would allow direct tests of which drug and class characteristics yield higher rebates: the number of close clinical substitutes in the class, whether generic or biosimilar entry has occurred, the share of class spending on protected sole-source products, and how concentrated the class is among a few manufacturers. The bargaining logic in Section 2.3 predicts larger pool effects where substitutes are many, while the alternative reading in Appendix F is that pooling's surplus concentrates in protected classes where the supplemental rebate is the only price lever, and only class-coded data can separate the two. Class-level work would be most useful for the high-cost classes where access matters most, such as biologics for autoimmune disease and targeted cancer drugs, since those are where a pool's bargaining position and a state's coverage decisions meet. Net spending effects are also highly desirable to estimate, although drug-level Medicaid rebate data are unlikely to be available to researchers given how closely they are held.

7. Conclusion

Two-thirds of U.S. states have joined an interstate Medicaid drug purchasing pool, yet robust evidence on whether pooling actually raises supplemental rebate capture is virtually nonexistent. Using a novel primary-source panel of pool membership for all 50 states from 2002 through 2024, and estimators matched to each pool's institutional structure, I find that joining a pool increases rebates from 1-2%. These findings are consistent with the theory of countervailing-buyer-power in which pooling increases bargaining power and ability to steer towards lower cost drugs. I caution against interpreting evidence as causal due to policy confounding and data issues (aggregation, reporting), my findings are consistent with a modest positive effect.

Pooling can be a useful fiscal tool to increase supplemental rebates, but the magnitude of its effects and ability to actually reduce drug spending are uncertain; and may also depend on the type of pool that is joined and other concurrent pharmacy policy changes. In a decade when federal law is reducing approximately a trillion dollars in Medicaid financing, pooled rebate negotiation is one of the few cost-containment levers that asks manufacturers, rather than enrollees or providers, to absorb a small part of the adjustment.

References

- Bendicksen, L., and Kesselheim, A.S. (2022). Anticipated efficiencies, real costs: Medicaid managed care organizations and the pharmacy benefit. *Journal of Managed Care & Specialty Pharmacy*, 28(3), 354-361.
- Borusyak, K., Jaravel, X., and Spiess, J. (2024). Revisiting event-study designs: Robust and efficient estimation. *Review of Economic Studies*, 91(6), 3253-3285.
- Callaway, B., and Sant'Anna, P.H.C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2), 200-230.
- Cameron, A.C., Gelbach, J.B., and Miller, D.L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3), 414-427.

Centers for Medicare & Medicaid Services (2026a). National Health Expenditure Accounts, Table 16: Retail prescription drugs expenditures; levels, percent change, and percent distribution, by source of funds, selected calendar years 1970-2024.

Centers for Medicare & Medicaid Services (2026b). Medicaid pharmacy supplemental rebate agreements (SRA), as of March 2026.

Chown, J., Dranove, D., Garthwaite, C., and Keener, J. (2019). The opportunities and limitations of monopsony power in healthcare: Evidence from the United States and Canada. NBER Working Paper 26122.

Congressional Budget Office (2025). Supplemental cost estimate: Public Law 119-21, Title VII, Finance, Subtitle B, Health, Chapter 1, Medicaid, as enacted on July 4, 2025. October 28, 2025.

de Chaisemartin, C., and D'Haultfoeuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9), 2964-2996.

Dranove, D., Ody, C., and Starc, A. (2021). A dose of managed care: Controlling drug spending in Medicaid. *American Economic Journal: Applied Economics*, 13(1), 170-197.

Dubois, P., Lefouili, Y., and Straub, S. (2021). Pooled procurement of drugs in low and middle income countries. *European Economic Review*, 132, 103655.

Duggan, M., and Scott Morton, F. (2006). The distortionary effects of government procurement: Evidence from Medicaid prescription drug purchasing. *Quarterly Journal of Economics*, 121(1), 1-30.

Gifford, K., Winter, A., Wiant, L., Dolan, R., Tian, M., and Garfield, R. (2020). How state Medicaid programs are managing prescription drug costs: Results from a state Medicaid pharmacy survey for state fiscal years 2019 and 2020. KFF and Health Management Associates, April 2020.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2), 254-277.

Hernandez, I., and Gellad, W.F. (2020). Differences between managed care and fee-for-service Medicaid in the use of generics for high-rebate drugs: The cases of insulin glargine and glatiramer. *Journal of Managed Care & Specialty Pharmacy*, 26(2), 154-159.

Ho, K., and Lee, R.S. (2024). Contracting over pharmaceutical formularies and rebates. NBER Working Paper 32790.

Horvath, J. (2019). State initiatives using purchasing power to achieve drug cost containment. National Academy for State Health Policy.

MacKinnon, J.G., and Webb, M.D. (2017). Wild bootstrap inference for wildly different cluster sizes. *Journal of Applied Econometrics*, 32(2), 233-254.

Medicaid and CHIP Payment and Access Commission (2018). Medicaid payment for outpatient prescription drugs. MACPAC Issue Brief, May 2018.

Medicaid and CHIP Payment and Access Commission (2026). MACStats: Medicaid and CHIP data book, February 2026, Exhibit 28: Medicaid gross spending and rebates for drugs by delivery system, FY 2024.

Munshi, K.D., Mager, D., Ward, K.M., Mischel, B., and Henderson, R.R. (2018). The effect of Florida Medicaid's state-mandated formulary provision on prescription drug use and health plan costs in a Medicaid managed care plan. *Journal of Managed Care & Specialty Pharmacy*, 24(2), 124-131.

Office of Inspector General, U.S. Department of Health and Human Services (2014). States' collection of offset and supplemental Medicaid rebates. Report OEI-03-12-00520, December 2014.

Parmaksiz, K., Pisani, E., Bal, R., and Kok, M.O. (2022). A systematic review of pooled procurement of medicines and vaccines: Identifying elements of success. *Globalization and Health*, 18(1), 59.

Rambachan, A., and Roth, J. (2023). A more credible approach to parallel trends. *Review of Economic Studies*, 90(5), 2555-2591.

Sun, L., and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2), 175-199.

Virabhak, S., and Shinogle, J.A. (2005). Physicians' prescribing responses to a restricted formulary: The impact of Medicaid preferred drug lists in Illinois and Louisiana. *American Journal of Managed Care*, 11(Spec No.), SP14-SP20.

Appendix

Appendix A. Additional Descriptive Material

Outcome reconstruction and reported-value anomalies. The analytic outcome was rebuilt directly from the per-fiscal-year sheets of the CMS-64 workbook, separating the FFS sidebar, managed-care sidebar, CHIP counterpart, and post-2020 opioid-treatment sidebar lines in the numerator from total drug spend in the denominator. The rebuilt series reproduces the analytic series to within rounding on all 1,104 state-years where both exist. Six state-years carry negative reported values: Hawaii 2017, North Carolina 2013, Virginia 2012, and Wisconsin in 2004, 2008, and 2010. CMS-64 reporting includes manufacturer credits and prior-period adjustments, so these likely reflect accounting adjustments and not errors, and they are retained without adjustment. Thirty-nine state-years are missing and are left as missing.

Exact zeros are common: 313 of the 1,111 analysis state-years report no supplemental collections at all, and Table 5 reports the zero shares by group. Movements in the capture rate therefore mix states that start reporting anything at all with states that deepen a rebate they already collect. The two-part analysis that would separate them, the probability that a state reports any positive collection and the capture rate conditional on positive reporting, is the natural next cut of this outcome, and Section 6.3 carries the corresponding limitation.

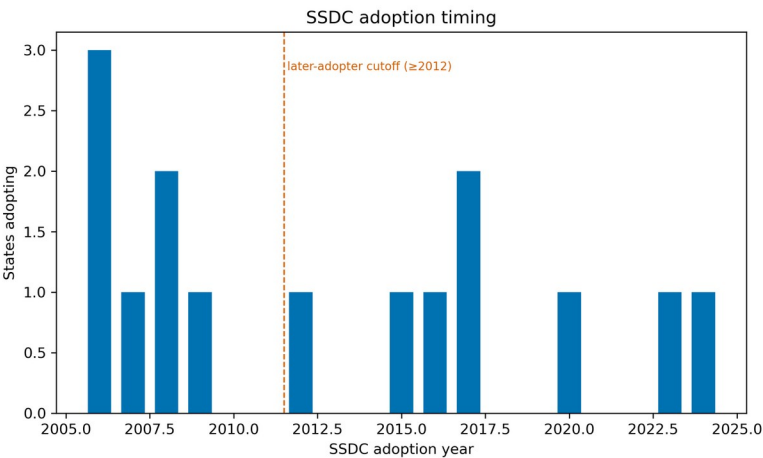
Table A1. Supplemental Rebate Capture by Pool Family, 2002-2024

Pool family	Mean	Median	SD	Min	Max	State-years
-------------	------	--------	----	-----	-----	-------------

SSDC	5.32%	4.83%	4.28%	0.00%	28.40%	181
TOP\$	3.52%	3.61%	2.56%	-0.08%	12.35%	149
NMPI	3.11%	2.20%	3.56%	-8.11%	28.24%	202
No pool	1.77%	0.16%	2.76%	-2.98%	21.83%	579

Notes: The unit of observation is the state-year, and pool family is membership in that year, so a state contributes to more than one row if it switched pools. The outcome is the supplemental rebate capture rate, supplemental rebate collections reported on the CMS-64 as a percentage of gross Medicaid drug expenditures. Negative values are reported CMS-64 values retained without adjustment (Section 3.1). These are raw cross-state comparisons and are not causal. The large mean-median gap for no-pool state-years reflects a right-skewed distribution with many states reporting near-zero collections. Author’s calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

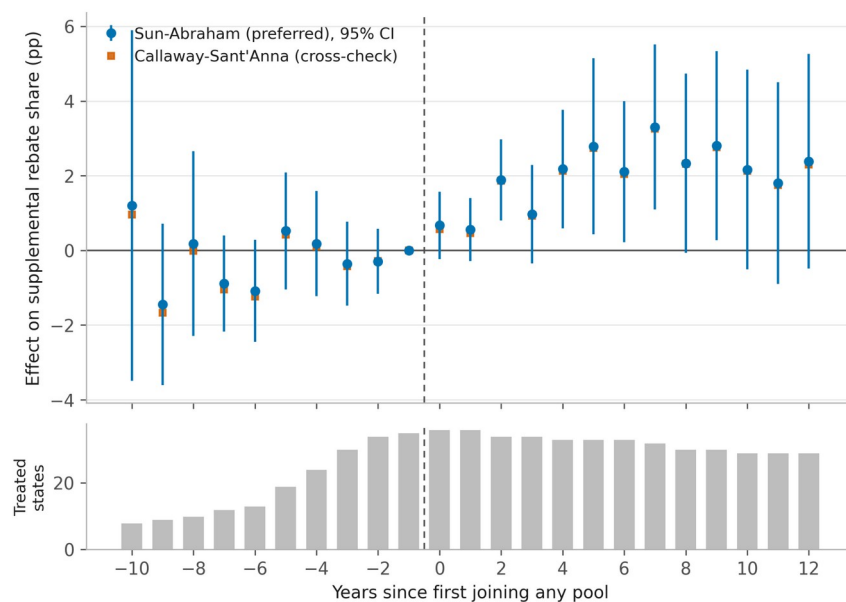
Figure A1. SSDC adoption timing



Notes: Number of states first adopting SSDC in each year, 2006-2024. The dashed line marks the 2012 cutoff separating the seven early adopters from the eight later adopters. Author’s coding from the primary-source pool-membership panel.

Appendix B. Multi-Pool Results and Master Robustness

Figure B1. Any-pool event study over the full event-time range



Notes: The Sun-Abraham interaction-weighted event-study coefficients and 95 percent confidence intervals plotted in Figure 3, with the Callaway-Sant’Anna cross-check, shown over the full plotted range of event times -10 through +12. Event time -1 is the omitted reference, standard errors are clustered on state, and the lower panel reports the number of treated states observed at each event time. The earliest leads are identified by as few as eight treated states, which is the sparsest region of the figure, while 29 of the 36 treated states are still observed at event time 12; reaching that horizon in a panel ending in 2024 requires joining by 2012, so the longest lags describe the earlier-adopting cohorts. The Sun-Abraham standard error rises from 1.226 pp at event time 8 to 1.465 pp at event time 12. Figure 3 restricts the window to event times -6 through +8. Author’s calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

Vendor-pool institutional detail. The two vendor-administered pools negotiate collective supplemental rebate agreements on behalf of member states. NMPI is administered by Magellan/Prime Therapeutics: the pool entity contracts directly with manufacturers, while member states choose which negotiated drugs to place on their own formularies, and its membership includes large states such as New York and Michigan. TOP\$ is likewise vendor-administered, through Prime/Provider Synergies. SSDC members instead join through a consortium agreement cost-shared through Vermont, the managing member, without contracting with the consortium’s hired contractor.

Table B1. Effect on Supplemental Rebate Capture by Pool Family

Treatment group	Treated states	Two-way fixed effects	WCB p	Sun-Abraham	Callaway-Sant’Anna	Exit-robust dCDH [95% CI]
Any pool	36	1.711	0.013	2.131**	2.006**	1.218 [0.12, 2.31]
		(0.664)		(1.005)	(0.906)	(0.559)
State-run pool	15	2.687	0.019	0.605	0.449	0.048 [-3.20,

(SSDC)						3.30]
		(1.030)		(1.877)	(1.609)	(1.658)
Vendor-run pools (NMPI or TOP\$)	26	1.184	0.073	2.110**	2.002*	1.363 [0.15, 2.58]
		(0.635)		(0.999)	(1.105)	(0.619)
NMPI alone	16	1.293	0.204	2.211*	2.109**	1.904 [0.01, 3.80]
		(0.866)		(1.154)	(1.052)	(0.965)
TOP\$ alone	11	1.592	0.057	1.632*	1.512	0.343 [-1.02, 1.71]
		(0.797)		(0.920)	(0.960)	(0.695)
SSDC later adopters (2012+)	8	3.705	0.051	3.376***	3.076***	1.972 [-0.15, 4.10]
		(1.492)		(0.769)	(0.848)	(1.084)
SSDC versus other-pool states	15	2.078	0.061	-0.270	-0.274	-0.270 [-3.55, 3.01]
		(0.952)		(1.532)	(1.472)	(1.675)

Notes: Each row is a separate estimation. Standard errors clustered on state are in parentheses below each point estimate. All specifications include state and year fixed effects and no other covariates, and states are weighted equally. Rows are ever-membership estimates and the categories overlap over time, because five states switched from a vendor pool into SSDC (Delaware, Kentucky, Pennsylvania, Vermont, and West Virginia), so a switching state contributes to more than one row at different dates. The comparison group is the same 13 never-pooled states in every row except the last, where the comparison group is the 21 states that were ever in a vendor pool but never in SSDC. “WCB p” is the wild cluster bootstrap p-value for the two-way fixed effects estimate. An alternative governance coding that groups SSDC and NMPI together as state-directed against TOP\$ alone as vendor-run gives 2.326 (SE 0.758) under two-way fixed effects, 2.038 under Sun-Abraham, and 1.906 under Callaway-Sant’Anna for the state-directed group. Sun-Abraham values reflect the implementation correction documented in Appendix G. Stars on the Sun-Abraham and Callaway-Sant’Anna columns are based on the analytic cluster-robust p-values and overstate significance with 13 never-pooled comparison states; the two-way fixed effects column carries its wild cluster bootstrap p and the dCDH column its bootstrapped interval, so those columns carry no markers. *p<0.10, **p<0.05, ***p<0.01. Author’s calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

Table B2. Current-Membership Estimates by Pool, Simultaneous Indicators

Current membership in	Two-way fixed effects	WCB p	Exit-robust dCDH [95% CI]
SSDC	2.887 (0.977)	0.012	-0.148 [-3.42, 3.13]
NMPI	1.462 (0.767)	0.100	1.271 [-0.55, 3.09]
TOP\$	1.072 (0.783)	0.220	0.101 [-1.19, 1.39]

Notes: The two-way fixed effects column comes from one model carrying three simultaneous current-membership indicators, so a state that moves from TOP\$ into SSDC contributes to both. The dCDH column estimates each pool’s current-membership effect separately against the never-pooled states. “WCB p” is the wild cluster bootstrap p-value for the two-way fixed effects estimate. Standard errors clustered on state in parentheses. Author’s calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

Absorbing versus current-membership treatment. Table 4 reports both definitions for the any-pool

treatment. The absorbing definition switches the indicator on at first entry and leaves it on, and gives 1.71

to 2.13 pp depending on estimator; the current-membership definition gives 1.83 pp under two-way fixed effects and 1.22 pp under dCDH. Six states leave a pool during the panel (Georgia, Hawaii, Nevada, New York, Pennsylvania, and Tennessee), contributing 48 of the 580 treated state-years under the absorbing definition. The absorbing definition identifies outcomes following first entry and is the only one the Sun-Abraham and Callaway-Sant'Anna estimators accept. Splitting the absorbing indicator into in-pool years and post-exit years gives +1.74 pp ($p = 0.011$) and -0.35 pp ($p = 0.78$), so the post-exit years are few and centered on zero and they dilute the average without distorting it. Estimating the exit side directly on the ever-pooled sample gives a Sun-Abraham effect of leaving a pool of -6.79 pp (SE 5.36, $p = 0.21$), which is correctly signed, entirely imprecise, and rests on six states.

Accounting, reporting, and counterfactual detail. This block carries the underlying numbers for the sensitivity blocks in Section 5.3, and Table B5 carries the rows not shown in Table 4.

The accounting convention matters mainly for the join year. Under the lagged denominator the event-time coefficient at entry falls from 0.70 to 0.18 pp. The any-pool event-time coefficients at event years three, five, and eight move only from 0.99, 2.84, and 2.47 to 0.85, 1.98, and 2.20, so the convention shifts the join year and leaves the ramp intact.

The result sits in the numerator. Modeling the two sides of the ratio separately, the inverse hyperbolic sine of supplemental collections rises by 2.00 (bootstrap $p = 0.009$). Log gross drug spending moves by -0.13 log points ($p = 0.32$), so the denominator does not respond to pool entry.

Timing variants move the estimate little. Dropping the partial entry year gives 2.07 pp, and lagging the treatment one or two years gives 1.79 and 1.97 pp.

The managed-care reporting margin is checked two ways. The managed-care sidebar column of the CMS-64 is populated for 30 of 49 states, first in 2012 and most commonly between 2013 and 2021. Restricting the numerator to the fee-for-service sidebar lines gives 1.91 pp (bootstrap $p = 0.016$). Dropping the two treated states whose managed-care line first appears within two years of entry gives 2.04 pp ($p = 0.006$).

Reweighting the comparison group pulls the estimate down. Entropy balancing on baseline collections and log drug spending gives 1.49 pp under two-way fixed effects and 1.52 pp under Sun-Abraham. Propensity-score weighting on those two moments plus baseline statutory-rebate share gives 1.43 and 1.37 pp. Two balanced moments are also the practical limit. No set of weights on the 12 usable donor states matches a third baseline moment, and near-solutions push nearly all of the weight onto a single state while balance gets worse than with no weighting at all. The never-pooled group supports only coarse reweighting toward the treated states' baseline profile, which is the caveat attached to these rows in Section 5.3.

The synthetic family lands nearby. Synthetic difference-in-differences run cohort by cohort against the never-pooled donors gives 1.88 pp on the 2002-2024 panel and 1.58 pp on the 1997-2024 extension. Synthetic control gives 1.35 and 1.43 pp, and staggered augmented synthetic control gives 1.09 pp (SE 0.83).

Extending the panel back to 1997 and re-running the primary specifications gives 1.59 pp under two-way fixed effects (bootstrap $p = 0.003$), 2.11 under Sun-Abraham, and 1.99 under Callaway-Sant'Anna. Dropping Tennessee gives 1.93 pp under Sun-Abraham alongside the 1.59 pp two-way fixed effects estimate reported in Table 4.

The reform audit behind the observed-reform exclusion rows is built from 50 documented reform events across 32 states in four families. Thirteen of the 36 treated states have no documented reform of any kind (Appendix I).

Later-adopter cutoff. The panel begins in 2002, so requiring at least K clean pre-entry years implies a cutoff of $2002 + K$, and the 2012 cutoff is the rule requiring at least ten pre-entry years. SSDC has no adoptions in 2010, 2011, 2013, or 2014, so the 2010, 2011, and 2012 cutoffs select the identical eight states: Delaware, Kentucky, Mississippi, North Dakota, Ohio, Oklahoma, Pennsylvania, and South

Dakota. The 2013 and 2014 cutoffs drop Mississippi, raise the point estimate to 3.93 pp, and halve the precision (bootstrap $p = 0.102$).

Later-adopter inference and the early-cohort null. Pool entry was not randomized, and without a defensible assignment model the reassignment exercise behind the 0.004 value in Table B4 is a permutation placebo and not design-based inference, so it rests on an exchangeability assumption the institutional record does not support and is not privileged over the other rows in that table. The point estimate is stable across implementations, samples, and donor groups (Table B3), and that stability does not offset the failure under family-wise correction and under any allowance for post-period differential trends. The later-adopter event study is plotted in Figure E1. Three explanations are available for the contrasting null among the seven early SSDC entrants. Their pre-periods are short and unequal, which is a mechanical problem. SSDC in 2006-2012 was also a small consortium of small states, so the early cohorts spent most of their observed post-period in a low-leverage regime. A third possibility is that the later joiners were simply better prepared, with the co-policies a pool needs already in place at entry, but the reform audit leans against that, since six of the eight entered without a documented reform within a year (Appendix I). The states with truncated pre-periods are exactly the states that were in the pool when it was small, so the first two explanations cannot be separated here.

Table B3. SSDC Later-Adopter Robustness Battery

Specification	Estimate and inference
Headline, Sun-Abraham, uncontrolled	3.38 pp, WCB $p = 0.051$
Two-way fixed effects	3.71 pp, WCB $p = 0.051$
Cross-estimator spread	Sun-Abraham 3.38; Callaway-Sant'Anna 3.08; Borusyak-Jaravel-Spiess 4.27; dCDH about 2 to 3.8; saturated event study 3.66
Cutoff sensitivity	2010+, 2011+, and 2012+ select the identical eight states (3.71 pp, WCB $p = 0.051$); 2013+ and 2014+ drop Mississippi (3.93 pp, WCB $p = 0.102$)
Leave-one-treated-state-out	[2.91, 3.88] pp
Leave-one-comparison-state-out	[3.12, 3.60] pp
Alternative donor pool (ever-vendor, never-SSDC)	3.41 pp

states)	
No-observed-reform subsample (six states, no reform within one year)	3.11 pp (SE 0.88)
Outliers dropped or winsorized at 1/99	3.07 to 3.38 pp
Outcome-construction covariates	3.6 to 4.0 pp, WCB p = 0.03 to 0.04
Spending-weighted	4.37 pp
Treatment lagged one year / entry year dropped	3.52 pp / 3.98 pp
Permutation placebo over adoption timings	placebo mean 0.04 pp (SD 1.20); two-sided p = 0.004
Rambachan-Roth sensitivity bounds	survives exactly parallel trends, [1.11, 5.49]; zero enters once any post-period violation is allowed
Governance-controlled sensitivity specification	sign-unstable and imprecise (Appendix D)
Pre-2002 panel extension	3.34 pp against 3.38 pp on the 2002-2024 panel

Notes: All rows use the eight SSDC states first treated in 2012 or later against the 13 never-pooled states, with state and year fixed effects and no covariates except where a row says otherwise. “WCB” is the wild cluster bootstrap. The pre-2002 extension parses the CMS-64 state sidebar rebate line, defined on the form in 1998, to build a 1997-2024 outcome panel; the 1997-2001 years add pre-period without adding treated mass, since every SSDC adoption is 2006 or later. The Callaway-Sant’Anna value of 3.08 supersedes an earlier 2.66 obtained when the group variable was passed to the estimation routine as an integer, which silently removed the never-treated group and substituted the last treated cohort as the comparison. Author’s calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

Table B4. Inference Procedures for the Later-SSDC Estimate

Procedure	Value	What it assumes or controls
Wild cluster bootstrap p	0.051	State-clustered finite-sample inference on a single hypothesis
Permutation-placebo p	0.004	Exchangeability of adoption timing across states, which is not demonstrated
Holm-adjusted p, five pool-definition hypotheses	0.154	Family-wise error control across the pool definitions tested
Benjamini-Hochberg adjusted p, same family	0.071	False-discovery control across the same family
Rambachan-Roth relative-magnitude breakdown	$\bar{M} = 0$	Zero enters the confidence set once any post-period trend violation is allowed

Notes: All rows describe the same estimate for the eight states first entering SSDC in 2012 or later against the 13 never-pooled states (469 state-years, 21 states). The two-way fixed effects point estimate is 3.705 pp and the Sun-Abraham point estimate is 3.376 pp. The permutation placebo draws 1,000 reassignments of the eight observed adoption timings to states in the analytic sample; its null distribution has mean 0.04 pp and standard deviation 1.20 pp. The hypothesis family is the five pool definitions estimated on the same panel with the same procedure: any pool, SSDC all adopters, NMPI, TOP\$, and SSDC later adopters. Author’s calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

Table B5. Additional Sensitivity Rows for the Any-Pool Estimate

Specification	Two-way fixed effects (pp)	WCB p	Heterogeneity-robust (pp)	N
Continuous membership, censored at	1.673	0.015	SA 2.252; CS 2.096	1,043

first exit				
Two-year moving collections-to-spending ratio	1.522	0.026	-	1,055
Partial entry year dropped	2.066	0.004	-	1,068
Treatment lagged one year	1.790	0.003	-	1,104
States with managed-care line onset near entry dropped	2.042	0.006	-	1,058
Propensity-score-weighted comparison group	1.433	0.024	SA 1.369	1,089
Synthetic DID, 1997-2024 panel	1.580	-	SC 1.427; ASCM 1.086	1,372
1997-2024 panel, primary specification	1.593	0.003	SA 2.110; CS 1.985	1,361
Observed-reform states dropped, reform within two years	1.849	0.019	SA 2.378	1,019

Notes: These rows continue Table 4 and use the same outcome, comparison group, fixed effects, and equal state weights. “WCB p” is the wild cluster bootstrap p-value for the two-way fixed effects estimate. SA is Sun-Abraham, CS is Callaway-Sant’Anna, SC is synthetic control, and ASCM is the staggered augmented synthetic control estimator. Sun-Abraham values reflect the implementation correction documented in Appendix G. The 1997-2024 panel admits all 15 cohorts to the synthetic-control family, while the 2002-2024 panel admits 13. Author’s calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

Appendix C. Pool-Membership Coding Caveats and Entry Timing

Eleven states carry a coding caveat. Tennessee is the only state whose full 2002-2024 sequence no second source corroborates and is coded provisionally as NMPI 2007-2010, TOP\$ 2011-2020, and no pool thereafter. Hawaii’s exit year rests on no located termination state plan amendment. New York’s 2013 re-entry could be 2012 and Georgia’s 2009 exit could be 2008. Pennsylvania’s TOP\$ start is coded 2007 rather than 2006. West Virginia, Utah, and South Dakota involve partial-year effective-date conventions, and Iowa and Maine are discussed below.

Entry dates behind the SSDC codings are documented as follows. Eleven of the fifteen SSDC states have a day-resolved documented effective date, and six of those eleven are mid-year, so the coded entry year contains a partial year of pooled rebates. Iowa and Maine have no documented effective date on file: four independent codings agree on SSDC from 2006, but no state plan amendment or effective date is recorded. Pennsylvania’s date is a state-fiscal-year attribution rather than a document-stamped day and is coded medium confidence. South Dakota is the one state where the coded year deliberately trails the documented effective date, since the state plan amendment took effect on October 1, 2022 and only a

partial quarter falls in 2022, so the state is coded from 2023. Ohio is the only state with a documented agreement anchor clearly separated from its effective date, with a supplemental rebate agreement entered October 1, 2016 and SSDC membership effective January 1, 2017. First bid-cycle dates are not documented on file for any SSDC state and should not be presented as recoverable without new primary-source collection. Five of the fifteen SSDC states (Vermont, West Virginia, Delaware, Pennsylvania, and Kentucky) entered SSDC from another pool, so for those states SSDC entry is a governance switch rather than a first exposure to pooling.

Appendix D. Governance-Controlled Sensitivity Specification

Re-estimating each design with time-varying governance controls (uniform PDL, pharmacy carve-out, and the major reform flag) yields sign-unstable and imprecise estimates across pools, which is the expected symptom of conditioning on a part-mediator on a restricted post-2010 sample (Section 4.2). These are reported only as a controlled residual association and are not the preferred estimate for any pool.

Appendix E. Secondary Outcomes

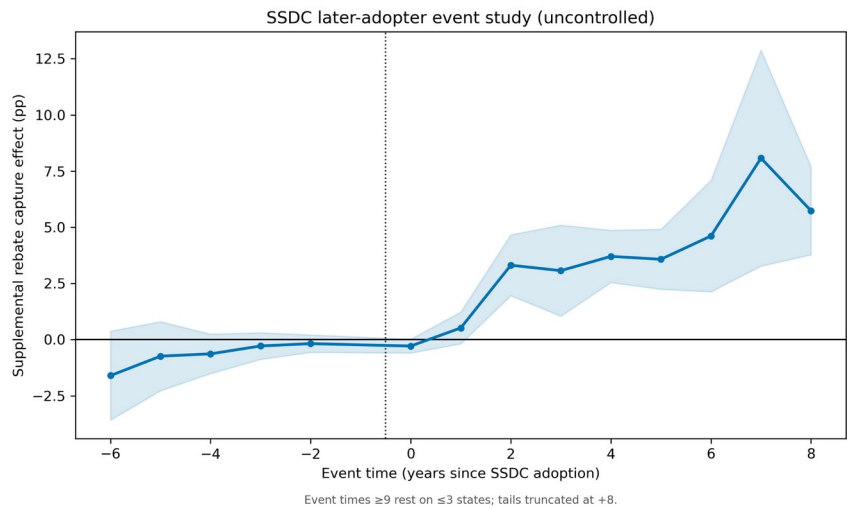
Table E1. Secondary Outcomes, SSDC Later-Adopter Sun-Abraham Estimates

Outcome	Effect	SE
Supplemental rebate capture (pp), primary outcome	3.38***	0.77
Log gross drug spending	-0.04	0.05
Log drug spending net of supplemental rebates (accounting translation)	-0.06	0.06
Supplemental rebate dollars per prescription	3.19***	1.04
Single-source spend share (pp)	-0.03	0.66
Innovator multiple-source spend share (pp)	-0.83	0.85
Top-10 NDC spend share (pp)	2.18***	0.69

Notes: Each row is a separate estimation using the Sun-Abraham interaction-weighted estimator on the eight SSDC later adopters against the 13 never-pooled states, with state and year fixed effects and no other covariates. Standard errors clustered on state. The net-of-supplemental-rebates row is gross spending times one minus the primary capture rate, so it is an accounting translation of the primary outcome rather than an independently measured fiscal outcome, and it cannot establish savings; the federally set national rebate is deliberately left in, for the reason given in Appendix J. Drug-mix and concentration outcomes come from State Drug Utilization Data and are most informative after 2010. The top-ten measure is based on spending. Enrollment-denominated per-capita outcomes are unavailable in the CMS-64 and are not estimated. Stars are

based on the analytic cluster-robust p-values and overstate significance with 13 never-pooled comparison states. *p<0.10, **p<0.05, ***p<0.01. Author’s calculations from CMS-64 Financial Management Report and State Drug Utilization Data.

Figure E1. SSDC later-adopter event study, supplemental rebate capture



Notes: Sun-Abraham interaction-weighted event-study coefficients with 95 percent confidence bands, for the eight states first entering SSDC in 2012 or later against the 13 never-pooled states. Event time -1 is the omitted reference. The path is truncated at event time 8 because event times of 9 or more are identified by three or fewer states. The effect is essentially zero in the entry window (event times 0 to 1, 0.2 pp), reaches 3.2 pp by years two to three, and 5.1 pp by years four to eight. Author’s calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

Appendix F. Drug Mix and the Class-Substitutability Test

Table F1. Drug-Mix Descriptives by Pool Family, SDUD, 2010-2024

Pool family	Single-source share	Innovator multiple-source share	Average spend per prescription	Top-10 NDC share
No pool	42.1%	21.6%	\$94.17	15.5%
SSDC	50.1%	21.4%	\$93.37	18.9%
NMPI	46.8%	20.5%	\$96.21	17.3%
TOP\$	43.4%	25.0%	\$94.64	16.6%

Notes: State-year means by pool family. Single-source products are protected brands with no generic equivalent. These descriptives motivate the composition checks in Section 5.6, which ask whether the measured collection gain could be a drug-mix artifact; the estimates there show that the single-source and innovator multiple-source shares do not move with pool entry. Author’s calculations from State Drug Utilization Data.

Table F1 reports the underlying drug-mix descriptives by pool family.

Class-substitutability test. The class-level prediction in Section 2.3 is that pooling should help most where close therapeutic substitutes make the steering threat valuable. A state-year proxy for

substitutability is the closest test these data allow, mirroring the size test with a baseline 2002-2005 median split and a wild cluster bootstrap on the high-versus-low differential.

Table F2. Class-Substitutability Differentials

Treatment	Proxy	High-minus-low differential	WCB p
SSDC later adopters	Innovator multiple-source share	-4.59 pp	0.13
SSDC all adopters	Innovator multiple-source share	-3.03 pp	0.13
Any pool	Innovator multiple-source share	-1.15 pp	0.30
SSDC or any pool	One minus single-source share	-0.9 to -0.02 pp	0.74 to 0.98

Notes: The differential is the pooling effect in high-substitutability states minus the effect in low-substitutability states, with states split at the median of the baseline 2002-2005 proxy value. Differentials are the interaction coefficients from two-way fixed effects models interacting the treatment indicator with the high-substitutability indicator, and “WCB p” is the wild cluster bootstrap p-value on that interaction, clustered on state. Author’s calculations from CMS-64 Financial Management Report and State Drug Utilization Data.

The cut does not confirm the prediction. The pool effect is no larger, and if anything mildly smaller, in more-substitutable states, and every differential is insignificant. Two caveats limit what this can establish. The proxy is a state-year drug mix rather than a within-class measure, so it confounds a state buying more substitutable drugs with correlated state characteristics. And a coherent alternative reading of the insignificant sign is that supplemental rebates are the only price lever available on protected sole-source drugs, so pooling’s surplus may concentrate where spending sits in those high-margin brands. A genuine test of the class-level mechanism requires a class-coded panel of the kind described in Section 6.4.

Appendix G. Estimation and Inference Details

Estimating equations. The static specification that produces the two-way fixed effects estimate in Table 4 is

$$Y_{st} = \alpha_s + \delta_t + \tau P_{st} + \varepsilon_{st}$$

where Y_{st} = the supplemental rebate capture rate, supplemental rebate collections reported as a share of gross drug spending, for state s in year t , α_s and δ_t = state and year fixed effects, P_{st} = an indicator equal to one if state s has entered a pool in year t or in any earlier year, τ = the average post-entry effect, and

ε_{st} represents an error term. Under the current-membership definition P_{st} instead equals one only in the years state s is in a pool.

The event-study specification replaces the single post indicator with a full set of event-time indicators,

$$Y_{st} = \alpha_s + \delta_t + \sum_{e \neq -1} \beta_e D_{st}^e + \varepsilon_{st}$$

where D_{st}^e = an indicator that state s is e years from its first year of pool membership, $e = -1$ = the omitted reference period, and β_e = the effect at event time e . Sun and Abraham (2021) estimate the β_e as an interaction-weighted average of cohort-specific effects, so that

$$\beta_e = \sum_g \omega_{g,e} \text{CATT}_{g,e}$$

where $\text{CATT}_{g,e}$ = the average effect at event time e for the cohort of states that first entered in year g , and $\omega_{g,e}$ = that cohort's share of the treated states observed at event time e . Each cohort is compared only against the never-pooled states, so no already-treated state is used as a comparison. The Sun-Abraham effect reported in the tables is the interaction-weighted average over all post-entry cohort-by-event-time cells: each event time contributes in proportion to the treated state-years observed there, over the full observed post-entry horizon. Table 4's common-window row instead restricts each estimator to event times 0 through 8, the window Figure 3 plots. For Sun-Abraham that is the unweighted average of the post-entry coefficients,

$$\beta^{\text{post}} = \frac{1}{9} \sum_{e=0}^8 \beta_e,$$

while the two-way fixed effects entry re-estimates the static model on the sample restricted to that window. The same equations define the SSDC later-adopter estimates, with the sample restricted to the eight states first treated in 2012 or later plus the thirteen never-pooled comparison states, and with D_{st}^e defined off the first SSDC year.

Sun-Abraham implementation. The Sun-Abraham estimates come from a fully saturated cohort-by-relative-time regression with event time -1 as the reference period, aggregated with the cohort-share weights above. An earlier software implementation excluded the single 2003-entry cohort from the interaction set, because that cohort's only pre-entry year is the reference period, while leaving its post-entry years in the estimation as comparison observations. The estimates it produced, including a 1.92 pp any-pool average, are superseded by the values reported here. The corrected implementation keeps the 2003 cohort, gives 2.13 pp, and matches an independent implementation in a second statistical package to four decimal places.

Exit-robust estimator. Because states enter and exit NMPI and TOP\$, the current-membership and pool-specific estimates use the de Chaisemartin and D'Haultfoeuille (2020) dynamic estimator on the full sample against never-pooled states, reporting the average effect, its bootstrapped confidence interval, and the placebo pre-period estimates. That estimator returns an entry (switchers-in) path here but cannot return a separate exit (switchers-out) path, because every treated state's first treatment change is an entry and period-one treatment is zero for all 49 states, so the design restriction the estimator requires for the exit path fails. The exit side is therefore estimated directly on the ever-pooled sample. Two-way fixed effects, where shown, is diagnosed with the Goodman-Bacon (2021) decomposition.

Wild cluster bootstrap. All headline inference uses the wild cluster bootstrap (Cameron, Gelbach, and Miller, 2008; MacKinnon and Webb, 2017) with Rademacher weights, 9,999 replications, clustered on state, with random seeds fixed for reproducibility. Every specification was re-estimated in a second statistical package as a check, and Table B3 reports two further implementations for the SSDC later-adopter sample.

Permutation placebo. For the SSDC later-adopter sample I reassign the eight observed adoption timings to random states in the analytic sample and recompute the Sun-Abraham average post-adoption effect, over 1,000 reassignments. Pool entry was not randomized and no assignment model is established here, so this is a permutation placebo rather than exact design-based inference: it describes how unusual the

observed estimate is relative to arbitrary reassignments of timing, under an exchangeability assumption the institutional record does not support. Its p-value is reported alongside the wild cluster bootstrap, Holm, Benjamini-Hochberg, and trend-sensitivity results in Table B4 and is not privileged over them.

Multiplicity. The hypothesis family for the Holm and Benjamini-Hochberg adjustments is the five pool-definition hypotheses estimated on the same panel with the same procedure: any pool (1.711, WCB p = 0.013), SSDC all adopters (2.687, 0.019), NMPI (1.293, 0.204), TOP\$ (1.592, 0.057), and SSDC later adopters (3.705, 0.051).

Sensitivity to differential trends. Sensitivity is assessed with the Rambachan and Roth (2023) framework, which asks how large a post-period violation of parallel trends, benchmarked against the pre-period, would have to be before the confidence set for the average post-adoption effect includes zero. It is applied to the primary any-pool Sun-Abraham event study (1,111 observations, four pre-period leads and nine post-period lags binned at the endpoints, average post-adoption effect 1.683 pp, average pre-period lead -0.306 pp) and to the SSDC later-adopter event study.

Appendix H. Heterogeneity by Program Size

Table H1. Pooling Effects by State Medicaid Program Size

Specification	Estimate (pp)	SE	WCB p
Binary split, SSDC later adopters: small states	3.37	-	-
Binary split, SSDC later adopters: large states	1.25	-	-
Binary split, SSDC later adopters: test of the small-large difference	-	-	0.33
Binary split, any pool: small states	2.52	-	-
Binary split, any pool: large states	1.02	-	-
Binary split, any pool: test of the small-large difference	-	-	0.14
Continuous interaction: any pool x baseline log drug spend	-0.253	0.407	0.565
Continuous model: effect at 25th percentile of size (about \$160M)	1.988	0.697	0.023
Continuous model: effect at mean size (about \$422M)	1.743	0.672	0.014

Continuous model: effect at 75th percentile of size (about \$887M)	1.555	0.798	0.088
--	-------	-------	-------

Notes: The binary split uses a median split on mean log Medicaid drug spending over 2002-2005 and reports Sun-Abraham estimates within each half, with the differential tested by wild cluster bootstrap. The continuous specification interacts the absorbing any-pool indicator with baseline mean log drug spending over 2002-2005, demeaned across states, in a two-way fixed effects model with state-clustered standard errors; the size main effect is absorbed by the state fixed effects. Each reported level effect is read off the treatment coefficient in a model re-centered at that size. The continuous specification loses Arizona, which has no 2002-2005 drug-spending values, leaving 1,089 state-years across 48 states. Moving from the 25th to the 75th percentile of baseline size is a 1.71 log-point increase and shifts the estimated pooling effect only from 1.99 to 1.56 pp. Author's calculations from CMS-64 Financial Management Report data and the primary-source pool-membership panel.

Point estimates are larger for smaller states, but neither the binary differential nor the continuous interaction is statistically distinguishable from zero. The stronger small-state contrast in the binary split is specific to the SSDC later-adopter subsample. The later adopters include Pennsylvania, Ohio, and Kentucky, three of the larger state Medicaid programs, and they show large gains, which is not in tension with the size prediction: the bargaining argument concerns a state's own size holding the pool constant, and the leverage a large state acquires by entering an already-large pool comes from the bloc it joins. The size and scale channels are separate, and this design identifies the first weakly and the second not at all.

Appendix I. Co-Timed Reform Audit

Table I1. Co-Timed Pharmacy-Benefit Reforms Around SSDC Entry

State	SSDC entry	Uniform PDL (MCO)	Carve-out	MCO rebate collection	Single PBM
Mississippi	2012	-	-	post (+2)	post (+12)
North Dakota	2015	-	post (+5)	post (+2)	-
Delaware	2016	-	-	same year	-
Ohio	2017	post (+3)	-	post (+3)	post (+5)
Oklahoma	2017	-	-	post (+7)	-
Pennsylvania	2020	same year	-	same year	-
South Dakota	2023	-	-	-	-
Kentucky	2024	pre (2021)	-	pre (2021)	pre (2021)

Notes: Each cell records the timing of a major pharmacy-benefit reform relative to the state's SSDC entry year. "post (+n)" means the reform arrived n years after entry, "pre (year)" means it arrived before entry in the year shown, "same year" means it coincided with entry, and "-" means no such reform is recorded during the panel. Author's coding of pharmacy-benefit reforms from primary-source state Medicaid materials.

Only two of the eight later SSDC adopters, Delaware in 2016 and Pennsylvania in 2020, entered in the same year as a major reform. Re-estimating on the six remaining states gives 3.11 pp (SE 0.88) against the full-sample 3.38, and dropping the three largest recent adopters instead gives 3.61 pp.

The same audit extended to all 36 treated states uses a reform panel of 50 documented events across 32 states in four families: a uniform PDL binding managed-care plans, a pharmacy carve-out, extension of supplemental rebate collection to managed-care utilization, and a move to a single state PBM. Thirteen of the 36 treated states have no documented reform of any kind. Dropping the one state with a documented same-year reform gives 1.90 pp (Table 4) and dropping the four states with a reform within two years gives 1.85 pp (Table B5). This audit reduces concern about four specified reform families among the treated states the panel can see. It is coded from 2010 onward for 45 of 50 states and only four of the 50 documented events fall in the 2000s, while 26 of the 36 treated states entered a pool before 2010, so for those states the audit under-detects co-timed reform and the exclusions cannot establish that the remaining entries were free of co-timed changes. Unmeasured operational changes remain unaddressed.

Appendix J. Reported Net Spending and the Statutory-Rebate Ramp

Section 5.6 nets out only the supplemental rebate when asking whether higher collections translate into lower net spending. The CMS-64 worksheet also reports a directly measured net series, gross drug spending minus all rebate offsets, for every state from 2002 through 2024. That series cannot be interpreted as a pool-specific effect in this design, because treated and comparison states experienced different statutory-rebate trajectories over the panel.

Table J1. Net-Spending Outcomes, SSDC Later-Adopter Sun-Abraham Estimates

Outcome	Effect	SE
Log gross drug spending (analytic series)	-0.041	0.051
Log net of supplemental rebates (Table E1)	-0.060	0.064
Log gross drug spending (reported CMS-64)	-0.039	0.050
Log net of all rebates (reported CMS-64)	-0.378***	0.073

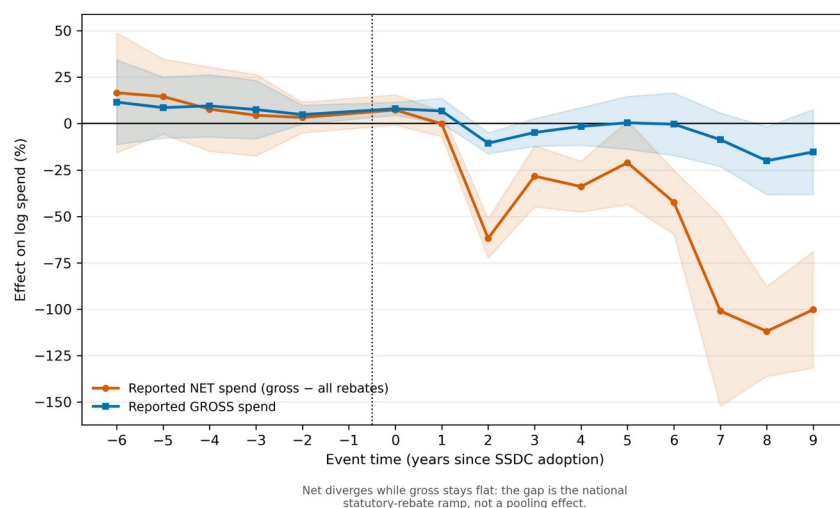
Notes: Each row is a separate estimation on the eight SSDC later adopters against the 13 never-pooled states, with state and year fixed effects. Standard errors clustered on state. Reported gross and net drug spending are extracted from the CMS-64 master worksheet. Stars are based on the analytic cluster-robust p-values and overstate significance with 13 never-pooled comparison states. *p<0.10, **p<0.05, ***p<0.01. Author's calculations from CMS-64 Financial Management Report data.

Table J1 reports four net- and gross-spending outcomes. The reported-gross estimate reproduces the analytic gross series, at -0.039 against -0.041, which confirms the extraction. The reported all-rebate net

swings to -0.378, an implausible 38 log point decline for a treatment that raises supplemental rebate capture by about 3.4 points. Three features explain it.

First, the timing is wrong. The event-time path of the reported net is flat and indistinguishable from zero through the adoption year, then falls discontinuously and keeps growing in magnitude, past -60 log points by event time 2 and beyond -100 at the longest lags, where cohort support is thinnest (Figure J1). An effect operating on a rebate margin would appear at adoption and scale with the roughly three points collected, instead of arriving with a two-year delay and compounding to a doubling of spend. The reported gross path stays flat throughout.

Figure J1. Reported CMS-64 net versus gross drug-spend event-time paths, SSDC later adopters



Notes: Sun-Abraham event-time coefficients for log reported net drug spending and log reported gross drug spending, eight SSDC later adopters against 13 never-pooled states. The window is truncated to event time -6 to +9 for legibility. Author's calculations from CMS-64 Financial Management Report data.

Second, the source of the gap is the statutory rebate rather than utilization. The implied total rebate share, one minus reported net over reported gross, rises from about 0.19 in 2002 to between 0.58 and 0.73 by 2024, reflecting the post-ACA escalation of inflation-indexed and best-price rebates plus the 2010 extension of rebates to managed-care utilization. This national ramp is common to treated and comparison states and is what year fixed effects exist to absorb.

Third, the ramp is differential across the comparison, which is why the year fixed effects do not absorb it. The mean total rebate share rises by roughly 0.55 for SSDC later adopters, from about 0.19 to 0.74, against roughly 0.39 for never-pooled states, from about 0.19 to 0.58. That is a gap of some sixteen points by 2024, five times the supplemental collections the pool delivers. Differences in drug mix and managed-care penetration make the federally set rebate accrue faster in the later-adopting states, and the reported net books that as a pool saving. Excluding the statutory component produces a cleaner supplemental-rebate accounting measure, though not necessarily a better fiscal or welfare outcome, and a decomposition of statutory-rebate exposure by drug would be needed to build one.

Appendix K. Bargaining Intuition and the Chain from Entry to Outcome

This appendix states the bargaining intuition summarized in Section 2.3 in compact form. It is an organizing device for the empirical predictions and not a derivation, and the assumptions it would need are listed at the end.

The setup. Consider a manufacturer negotiating with a state, or with a pool of states, over preferred status for its drug in some therapeutic class. The value to the manufacturer of winning preferred status in state s is

$$v_s = q_s m$$

where v_s = the value of preferred status in state s , q_s = the steerable volume that a preferred-status decision in state s moves toward the drug, and m = the manufacturer's margin on that volume. A single state bargaining alone splits v_s with the manufacturer according to their relative bargaining weights, and the supplemental rebate is the manufacturer's payment for avoiding the disagreement outcome of nonpreferred placement and lost volume. Writing θ for the states' bargaining weight, the rebate a state negotiating alone extracts is $R_s = \theta v_s$. When a set of states P pools, and if the manufacturer's disagreement outcome is losing preferred placement across the whole bloc, it bargains against a combined prize

$$V_P = \sum_{s \in P} v_s = m \sum_{s \in P} q_s$$

with an equilibrium rebate $R_P = \theta V_P$.

What this does and does not establish. The setup organizes the empirical predictions in Section 2.3: the prize aggregates steerable volume q_s rather than covered lives, so nominal enrollment without coordinated formulary action adds little; the smallest single-state prizes are the ones for which joining changes the outside option most; and the prize is largest where close therapeutic substitutes exist, since q_s is small in protected or sole-source classes. It does not by itself imply a higher per-unit rebate, a superproportional gain, or a small-state advantage. A formal version would have to specify the bargaining protocol, the commitment device that lets members act as one buyer, the rule allocating the pool's surplus across members, and each party's outside option, and would then have to show which assumption raises the pool's bargaining weight or makes the disagreement payoff nonadditive. The institutional description in Section 2.3 does not supply the central one: each member state retains its own PDL and its own final acceptance decision, so a manufacturer that fails to reach terms with one state does not necessarily lose the entire bloc. The predictions are therefore treated as hypotheses throughout, and the size prediction is tested and not identified (Appendix H).

Pool-type detail. Institutional features sharpen the state-owned-versus-vendor prediction. Property-rights logic predicts higher bargaining effort and better adaptation when the party that bears the budget consequence also controls the decisions, as in SSDC. State-specific supplemental rebate agreements also avoid the lowest-common-denominator problem that a single uniform vendor contract can create, and an annual member-state meeting builds a repeated relationship, so manufacturers face the same coordinated buyers year after year, which raises the expected cost of an unattractive opening offer. Section 5.4 reports that the data do not confirm this prediction.

Interference and treatment versions. The designs in this paper treat each state's entry as its own treatment and the never-pooled states as an untouched comparison, and the bargaining logic above strains

that assumption in two ways. A new member adds covered lives to the bloc, so its entry plausibly changes what the incumbent members can extract, and joining a small consortium in 2006 is a different treatment from joining an established one in 2020. Neither possibility is testable within the state-level designs here, and both point toward the pool-level design described in Section 6.4.

The chain from entry to outcome. The implied chain runs from a state's pre-treatment characteristics, to the decision to enter a pool, through a set of post-entry governance and contracting changes, to the measured outcome. Pre-treatment confounders sit upstream of pool entry and are valid to adjust for if measured at baseline: baseline collections, Medicaid program size, managed-care penetration, fiscal stress, administrative capacity, drug mix, and the political environment all plausibly drive both who pools and how much a state collects. The treatment node is pool entry itself. Downstream of pool entry sit the mediators, which are a unified PDL, a pharmacy carve-out, expansion of managed-care rebate collection, formulary steering, and vendor negotiation. These are the channels through which pooling is supposed to work and must not be conditioned on as if they were confounders. The outcomes are the capture rate, measured here, and, further downstream and not measured in this paper, net drug spending and utilization or access. Adjusting for a baseline characteristic that predates pool entry removes a genuine confounder; conditioning on a post-entry mediator opens the bad-control problem described in Section 4.2. This classification is the basis for the symmetric control policy in Section 4.2 and the sensitivity-only treatment of governance controls in Appendix D.